

Multilingual AI-Powered Government Revenue Services Platform

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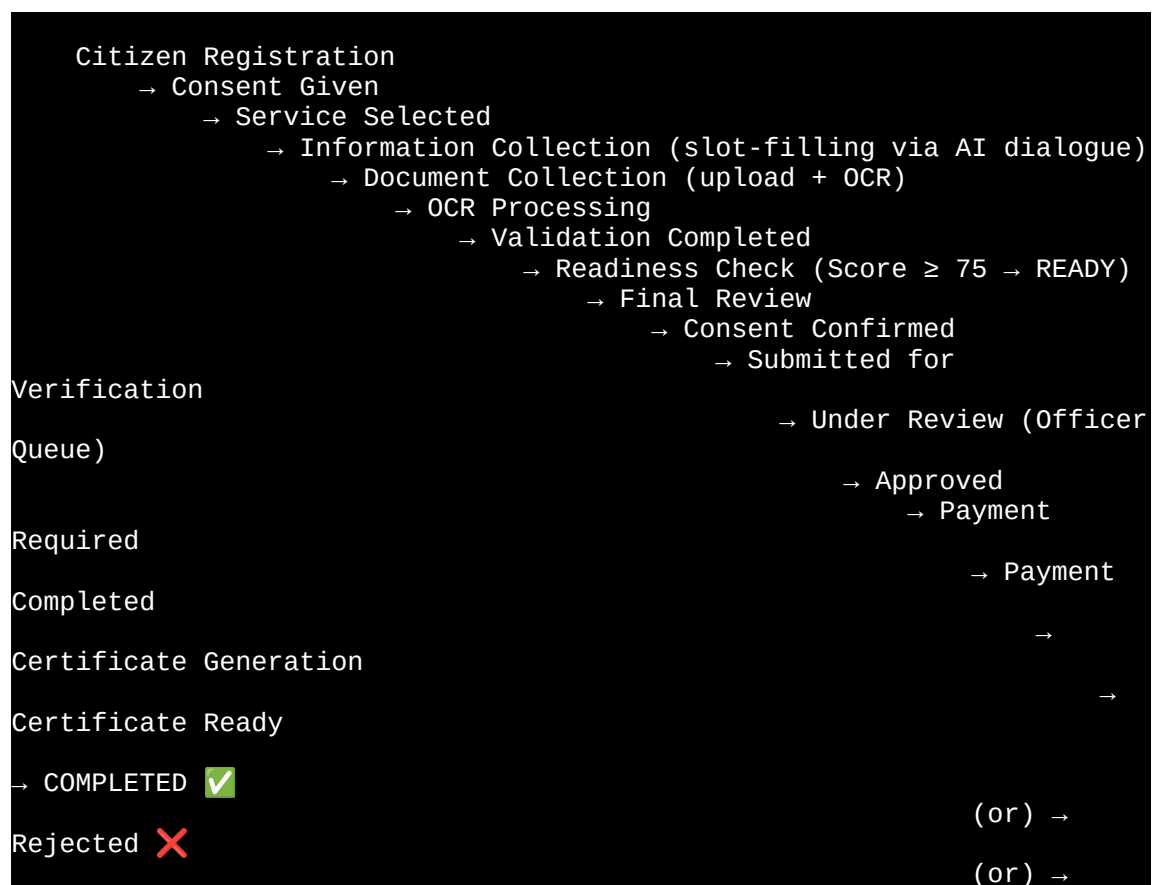
01. EXECUTIVE SUMMARY

RevenueGov (Rev_GOV) is an enterprise-grade, AI-powered government revenue services platform built to digitise and automate the entire lifecycle of citizen certificate applications across Indian state revenue departments. The platform currently supports four certificate types — **Income Certificate**, **Caste Certificate**, **OBC-NCL Certificate**, and **Domicile Certificate** — and is designed to be extended to additional services by simply adding a new YAML configuration file, with no new Python code required.

The platform is built on a **dual-engine architecture** that separates deterministic, auditable business logic from probabilistic AI inference. All eligibility checks, fee calculations, field validation, OCR extraction, and readiness scoring are executed deterministically using rule configurations loaded from YAML service specifications. The AI/LLM layer (powered by OpenRouter, Gemini, or Groq) is used exclusively for generating contextually appropriate conversational dialogue and optionally normalising raw OCR output — and it is **always filtered by the Data Guard PII Firewall** before any data touches an external API.

Complete Application Lifecycle

The platform takes a citizen from first contact to certificate delivery through a **21-state Finite State Machine (FSM)** that enforces strict, unidirectional, auditable state progressions:



Key Platform Statistics (from source):

- 4 certificate service types (YAML-driven, extensible)
 - 4 input channels: Web Chat, WhatsApp Clone, IVR Simulator, REST API
 - 21 FSM states with deterministic transition validation
 - 15 database tables with immutable audit trail
 - 23 automated test files covering 180+ test scenarios
 - 3 LLM providers supported (OpenRouter, Gemini, Groq)
 - 11 OCR field extractors (Aadhaar, PAN, DOB, Name, Income, etc.)
 - 5-component Readiness Engine with weighted scoring (0–100)
 - AES-256-GCM field-level encryption for all restricted PII
 - Zero PII ever transmitted to external LLM providers (Data Guard enforced)
-

02. PROJECT INTRODUCTION

Government certificate issuance in India traditionally involves multiple physical visits to revenue offices, manual form filling, document photocopying, and waiting days or weeks for officer review. The process lacks transparency, is susceptible to human error in data entry, and offers no real-time status visibility to citizens.

Rev_GOV solves these problems through an AI-first, digital-first platform that meets citizens where they are — whether on a modern web browser, WhatsApp, or a basic telephone call. The platform provides:

For Citizens:

- Conversational AI guidance through every step of the application, in their preferred language (English, Hindi, Marathi)
- Automatic OCR extraction of document data (Aadhaar, PAN, income certificates)
- Real-time application status updates via push notifications (SSE)
- Multi-channel continuity: start on WhatsApp, continue on web, check status via IVR
- Full transparency of the application process and required documents

For Government Officers / Admins:

- Centralised application review queue with filtering, search, and pagination
- AI-computed readiness scores and document match scores for each application
- One-click decision panel: Approve, Reject, or Request Clarification
- Complete, tamper-evident audit trail of all decisions and data changes
- Real-time dashboard with application statistics and pipeline analytics

Technical Value:

- Zero vendor lock-in for AI: switch between OpenRouter, Gemini, or Groq with a single `.env` change
- Zero manual code changes required to add new certificate services
- Production-ready security: JWT authentication, AES-256-GCM encryption, Data Guard PII firewall, rate limiting
- Fully testable: 23 test files with unit, integration, E2E, and security tests

03. PROBLEM STATEMENT & HOW REV_GOV SOLVES IT

The following table maps each real-world problem in government certificate processing to the specific technical mechanism in Rev_GOV that addresses it:

Problem	Rev_GOV Solution	Source Reference
Manual data entry errors	AI-guided slot-filling dialogue with immediate field validation	<code>rules_engine/engine.py</code> <code>FieldValidator</code>
Duplicate applications across channels	<code>CitizenResolver</code> with SHA-256 channel identity hashing	<code>data_layer/repositories/citizen_repo.py</code>
Document verification is slow and error-prone	Tesseract OCR + PyMuPDF with 10 deterministic field extractors	<code>services/ocr_service.py</code>
No PII protection when using AI	Data Guard blocks any RESTRICTED field from reaching LLM	<code>data_guard/guard.py</code> <code>DataGuard.check()</code>

Problem	Rev_GOV Solution	Source Reference
Lack of real-time status visibility	SSE EventBus pushes updates to citizen's browser the moment admin makes a decision	api/routes/stream.py EventBus
Inconsistent validation across services	Declarative YAML service specs with eligibility rules, fee calculations, and required documents	seed/service_specs/*.yaml
Fragmented multi-channel workflows	ConversationSession keyed by citizen_ref (not channel ID) provides cross-channel continuity	db_models.py ConversationSession
Payment before government approval	FSM + API guard: PAYMENT_REQUIRED state is ONLY reachable from APPROVED	application_fsm.py, payment.py
No audit trail of officer decisions	Append-only AuditLog table with blockchain-style hash chaining	db_models.py AuditLog
Inaccessible for low-literacy users	IVR channel with DTMF input and TTS audio responses	api/routes/ivr.py
Inconsistent fee calculations	Fee Calculator reads waiver conditions from YAML; no hardcoded fee amounts in Python	rules_engine/engine.py FeeCalculator
OCR noise corrupts extracted names	Deterministic name normaliser strips noise prefixes, OCR artifacts, stop-words	ocr_service.py _normalize_name()

04. OBJECTIVES & SCOPE

4.1 Primary Objectives

1. **Full digitisation** of the certificate application process from citizen registration to certificate delivery
2. **AI-powered guidance** that assists citizens through complex application forms in their preferred language
3. **Deterministic security**: all business rules (eligibility, fees, validation) enforced by the rules engine, never by the LLM

4. **PII protection by design:** the Data Guard ensures no personally identifiable information reaches external AI APIs

5. **Complete auditability:** every state change, field update, document upload, and admin decision is logged immutably

6. **Omnichannel continuity:** a citizen's application context persists across Web, WhatsApp, and IVR channels

4.2 Scope (Verified from Source Code)

In Scope:

- Citizen registration, authentication, and profile management
- Four certificate services: Income, Caste, OBC-NCL, Domicile
- Conversational AI dialogue for slot-filling (Web, WhatsApp, IVR)
- Document upload, OCR extraction, and deterministic field matching
- 21-state application lifecycle with FSM enforcement
- Admin review panel with approve/reject/clarification decisions
- Payment processing with OCR-based receipt verification
- PDF certificate generation and download
- Real-time push notifications (SSE)
- Complete audit trail with hash-chained log entries
- Multilingual support (English, Hindi, Marathi)

Out of Scope (Not verified in source code):

- Production deployment to cloud infrastructure
- Real payment gateway integration (currently using MOCK_AUTO mode)
- Real government API integration (currently using mock government adapter)
- SMS/WhatsApp notification via Twilio or similar (currently using mock notification adapter)

05. KEY FEATURES

5.1 AI-Powered Conversational Slot-Filling

The platform uses a `ConversationOrchestrator` (source: `backend/app/orchestration/state_machine/orchestrator.py`) that drives a progressive conversation to collect all required fields. Each message is processed through:

1. **NLU Intent Classifier** — detects intent (answer, greeting, help, question, channel switch)
2. **Field Validator** — validates the submitted value against the service spec's validation rules
3. **Next Question Engine** — determines which required field to ask for next
4. **LLM Dialogue Generator** — generates a contextually appropriate response (Data Guard filtered)
5. **Session Persistence** — saves collected fields to `ConversationSession.filled_slots`

5.2 Tesseract OCR with Deterministic Field Extraction

The `OCRService` (702 lines) extracts 10+ structured fields from uploaded documents using:

- **PyMuPDF** for text-native PDFs
- **Tesseract OCR** (OEM 3, PSM 6) for images and scanned PDFs
- **10 regex patterns** for deterministic field extraction (Aadhaar, PAN, DOB, Name, Income, Gender, Employer, Address, Transaction ID, Caste Category)
- **Deterministic normalisation** for each field type (name, date, income, Aadhaar, PAN, gender)
- **Optional AI normalisation** via OpenRouter (only after Data Guard approval with `SYNTHETIC` classification)

5.3 Data Guard PII Firewall

The `DataGuard` class (source: `data_guard/guard.py`) intercepts ALL outbound data destined for external LLM APIs. It:

- Scans all dict keys and values recursively for `RESTRICTED_FIELDS` (18 fields: Aadhaar, PAN, name, DOB, phone, etc.)
- Blocks payloads containing Aadhaar pattern `\b\d{4}\s?\d{4}\s?\d{4}\b` via regex

- Allows `QUASI_IDENTIFIER` fields only when marked `SYNTHETIC` and count < 3 (k-anonymity)
- Writes every `BLOCK` decision to the immutable `AuditLog` table
- Cannot be bypassed — `DataGuardBlockedError` stops the entire operation

5.4 5-Component Readiness Engine

Before a citizen can submit their application, the `ReadinessEngine` computes a weighted score (0–100) across five components. Score ≥ 75 AND zero blocking issues = `can_submit: true`.

5.5 21-State FSM with Strict Transition Validation

The `ApplicationFSM` enforces that applications can only move to valid next states. The most critical rule — **payment is only reachable AFTER government approval** — is enforced both in the FSM (`VALID_TRANSITIONS`) and at the API layer (`payment.py` guard).

5.6 Omnichannel Context Vault

Every citizen gets a single `ConversationSession` record keyed by their `citizen_ref` token. This session is shared across all channels. Switching from WhatsApp to Web does not lose any collected data.

5.7 Blockchain-Style Audit Trail

The `AuditLog` table uses an append-only pattern where each entry's `previous_hash` column references the SHA-256 hash of the preceding entry, creating a tamper-evident chain of custody that can be cryptographically verified.

06. FUNCTIONAL REQUIREMENTS

ID	Requirement	Implemented In
FR-01	Citizens can register with email or phone	api/routes/auth.py register_citizen()
FR-02	Citizens can log in and receive a JWT token	api/routes/auth.py citizen_login()
FR-03	Citizens can initiate applications via chat	api/routes/ conversation.py /message
FR-04	System collects required fields via AI dialogue	orchestration/ state_machine/ orchestrator.py
FR-05	Citizens can upload documents (images, PDFs)	api/routes/ conversation.py /document-upload
FR-06	System performs OCR on uploaded documents	services/ocr_service.py run_ocr()
FR-07	OCR results are matched against declared fields	services/ matching_service.py
FR-08	Mismatches are flagged and resolvable by citizen	api/routes/ conversation.py /resolve- mismatch
FR-09	Readiness score is computed before submission	services/ readiness_engine.py ReadinessEngine.compute()
FR-10	Application can be submitted when score ≥ 75	api/routes/ applications.py readiness guard
FR-11	Admin can view all submitted applications	api/routes/ applications.py /admin/list
FR-12	Admin can approve, reject, or request clarification	api/routes/ applications.py /admin/{id}/decision
FR-13	Citizen receives real-time notification of admin decisions	api/routes/stream.py SSE EventBus
FR-14	Payment can only be initiated after government approval	api/routes/payment.py state guard

ID	Requirement	Implemented In
FR-15	Payment receipt is OCR-verified	services/ payment_service.py
FR-16	Certificate PDF is generated after payment	services/ certificate_service.py
FR-17	Citizen can download issued certificate	api/routes/ applications.py /{id}/certificate
FR-18	Public tracking of application status	api/routes/tracking.py
FR-19	WhatsApp channel processes messages and documents	api/routes/whatsapp.py
FR-20	IVR channel accepts DTMF input and produces TTS	api/routes/ivr.py
FR-21	Citizen can switch channels without losing context	api/routes/ conversation.py /channel-switch
FR-22	Admin can view the audit log	api/routes/dashboard.py

07. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement	Implementation
Security	All restricted PII encrypted at rest	AES-256-GCM FieldEncryptor
Security	No PII sent to external LLM APIs	DataGuard.check() hard block
Security	JWT tokens expire after 8 hours	JWT_ACCESS_TOKEN_EXPIRE _MINUTES=480 in config.py
Security	Rate limiting on all API endpoints	SecurityMiddleware in core/security.py
Security	Citizen A cannot access Citizen B's data	citizen_ref ownership check on every query

Category	Requirement	Implementation
Auditability	All decisions persisted immutably	Append-only <code>AuditLog</code> with hash chain
Privacy	Channel identities stored as SHA-256 hashes	<code>channel_identities.identifier_hash</code>
Reliability	LLM failure returns HTTP 503, not silent fallback	<code>LLMUnavailableError</code> global handler
Extensibility	New service = new YAML file only	<code>ServiceSpecLoader.load_all()</code> scans directory
Performance	Knowledge base loaded once into memory	<code>RAGService</code> class-level <code>_chunks</code> cache
Performance	Service specs loaded once into memory	<code>ServiceSpecLoader._cache</code> dict
Portability	SQLite as default database	<code>DATABASE_URL=sqlite:///... in .env</code>
Observability	Structured logging with timestamps	<code>structlog==24.4.0</code>
Correctness	All state transitions validated by FSM	<code>ApplicationFSM.can_transition_to()</code>

08. TECHNOLOGY STACK

All packages verified from `backend/requirements.txt` and `frontend/package.json`.

8.1 Backend Stack

BACKEND STACK		
Layer	Technology	Version
Web Framework	FastAPI	0.115.0

ASGI Server	Uvicorn (standard)	0.30.6
ORM	SQLAlchemy	2.0.35
Database	SQLite (aiosqlite)	0.20.0
Validation	Pydantic	2.9.2
Settings	pydantic-settings	2.5.2
JWT Auth	python-jose	3.3.0
Password Hashing	passlib[bcrypt]	1.7.4
Encryption	cryptography	43.0.1
Config Parsing	PyYAML	6.0.2
Environment	python-dotenv	1.0.1
HTTP Client	httpx	0.27.2
Gemini LLM	google-generativeai	≥0.8.0
Language Detection	langdetect	1.0.9
Image Processing	Pillow	10.4.0
OCR Engine	pytesseract	≥0.3.10
PDF Processing	PyMuPDF (fitz)	≥1.24.0
Async Support	anyio	4.6.0
Testing	pytest	8.3.3
Async Testing	pytest-asyncio	0.24.0
Structured Logging	structlog	24.4.0

8.2 Frontend Stack

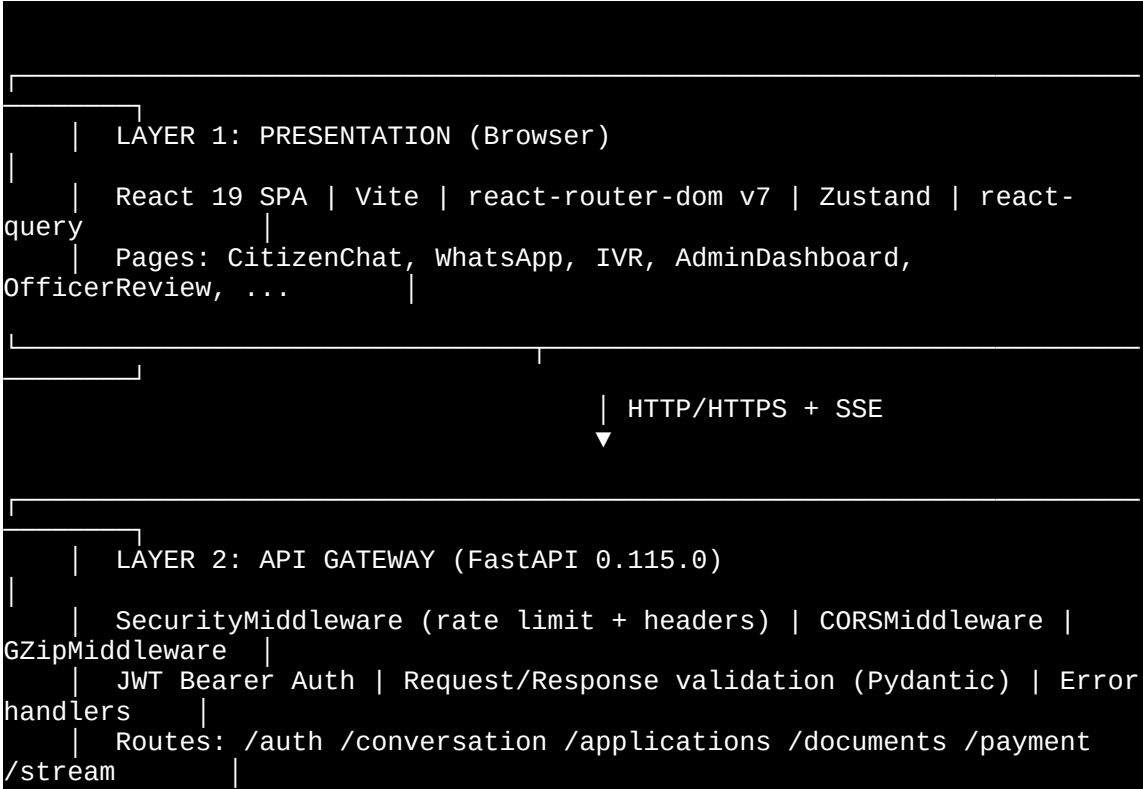
FRONTEND STACK		
Layer	Technology	Version
UI Framework	React	19.2.8
Build Tool	Vite	6.4.3
Router	react-router-dom	7.18.2
State Management	Zustand	5.0.15
Data Fetching	@tanstack/react-query	5.102.0
HTTP Client	axios	1.19.0
Animations	framer-motion	13.1.1
Charts	recharts	3.10.1
Icons	lucide-react	1.33.0
File Upload	react-dropzone	20.1.1
Markdown	react-markdown	10.1.0
Toast Notifications	react-hot-toast	2.6.0
Date Library	dayjs	1.11.23
CSS Utilities	clsx	2.1.1
Linters	oxlint	1.75.0

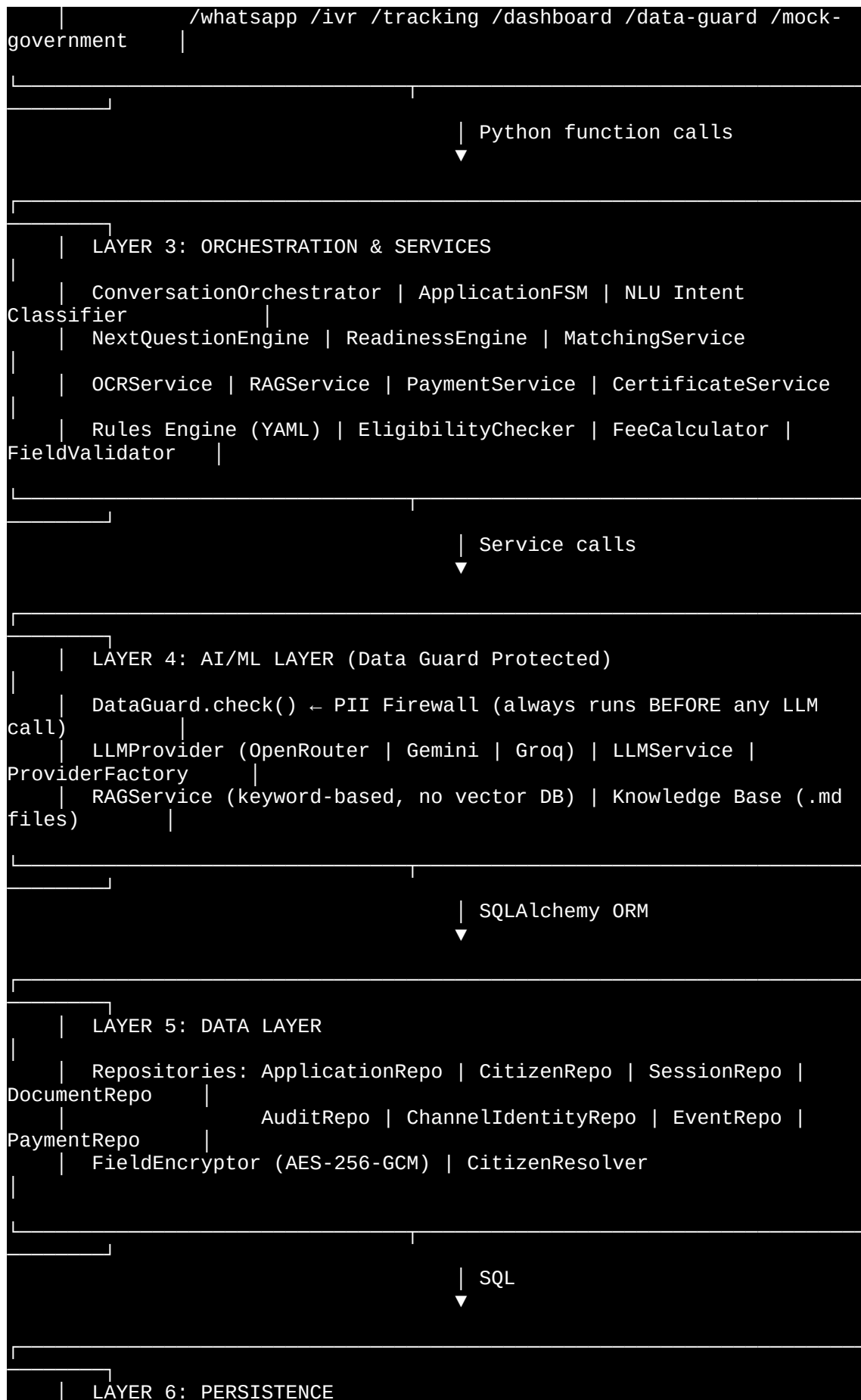
8.3 External Integrations

Service	Purpose	Configuration
OpenRouter API	Primary LLM provider (200+ models, free tier available)	OPENROUTER_API_KEY + OPENROUTER_MODEL
Google Gemini API	Alternative LLM provider	GEMINI_API_KEY + GEMINI_MODEL
Groq API	High-speed alternative LLM (Llama-3)	GROQ_API_KEY + GROQ_MODEL
Tesseract OCR	Local OCR engine for document extraction	TESSERACT_CMD path in .env
PyMuPDF	PDF text extraction (no external dependency)	pip package PyMuPDF

09. OVERALL SYSTEM ARCHITECTURE

The platform follows a **6-layer architecture** with strict separation of concerns:





```
|
|   | SQLite (revenue_services.db) | Local Filesystem (data/uploads,
data/certs) |
|   | YAML Service Specs | Markdown Knowledge Base
|
|_____
|_____
```

10. HIGH-LEVEL ARCHITECTURE

10.1 Component Interaction Diagram

```
graph TD
    CITIZEN["👤 Citizen<br/>(Browser/WhatsApp/Phone)"]
    ADMIN["🏛️ Admin/Officer<br/>(Browser)"]

    subgraph FRONTEND["Frontend (React 19 + Vite)"]
        CC["CitizenChat"]
        WA["WhatsApp UI"]
        IVR["IVR Simulator"]
        DASH["Admin Dashboard"]
        REV["Officer Review"]
        DGD["Data Guard Demo"]
    end

    subgraph API["FastAPI API Layer"]
        CONV["/conversation"]
        APPS["/applications"]
        AUTH["/auth"]
        PAY["/payment"]
        SSE["/stream (SSE)"]
        DOCS["/documents"]
        GOV["/mock-government"]
    end

    subgraph ORCH["Orchestration Layer"]
        ORC["ConversationOrchestrator"]
        FSM["ApplicationFSM (21 states)"]
        NQE["NextQuestionEngine"]
    end

    subgraph SERVICES["Services"]
        OCR["OCRService<br/>(Tesseract + PyMuPDF)"]
        RE["ReadinessEngine<br/>(5 components)"]
        MS["MatchingService"]
        RES["Rules Engine<br/>(YAML specs)"]
        PS["PaymentService"]
    end
```

```

    CS["CertificateService"]
end

subgraph AI["AI / LLM Layer (Data Guard Protected)"]
    DG["🛡️ DataGuard<br/>(PII Firewall)"]
    LLM["LLM Provider<br/>(OpenRouter/Gemini/Groq)"]
    RAG["RAGService<br/>(Keyword-based)"]
    KB["Knowledge Base<br/>(.md files)"]
end

subgraph DATA["Data Layer"]
    REPO["Repositories<br/>(Application, Citizen,<br/>Session,
Document, Audit)"]
    ENC["FieldEncryptor<br/>(AES-256-GCM)"]
end

DB[("SQLite<br/>revenue_services.db")]
FS["📁 Local Filesystem<br/>(uploads, certs, audio)"]
EB["EventBus<br/>(SSE Publisher)"]

CITIZEN --> CC & WA & IVR
ADMIN --> DASH & REV & DGD

CC & WA & IVR --> CONV & APPS & AUTH & PAY & DOCS
DASH & REV & DGD --> APPS & AUTH & GOV & SSE

CONV --> ORC
ORC --> FSM & NQE
ORC --> DG
DG -->|"ALLOWED"| LLM
DG -->|"BLOCKED"| ORC
LLM --> RAG
RAG --> KB

APPS --> RE & FSM
DOCS --> OCR
OCR --> MS
MS --> RE
RE --> RES

PAY --> PS
PS --> OCR
CS --> FS

APPS & GOV --> EB
EB --> SSE
SSE --> CC & DASH

ORC & APPS --> REPO
REPO --> ENC
REPO --> DB

style DG fill:#ff6b6b,color:#fff
style LLM fill:#4dabf7,color:#fff
style FSM fill:#51cf66,color:#fff
style DB fill:#fcc419,color:#000

```

10.2 Component Descriptions

Component	Description	Source File
ConversationOrchestrator	Central coordinator: receives citizen messages, routes to FSM, calls validators, drives LLM dialogue	orchestration/ state_machine/ orchestrator.py
ApplicationFSM	Enforces 21-state lifecycle. Only allows valid transitions. Contains multilingual citizen messages	orchestration/ state_machine/ application_fsm.py
DataGuard	PII firewall. Scans all data destined for LLM. Blocks RESTRICTED fields. Logs every decision	data_guard/ guard.py
OCRService	Extracts text from documents using Tesseract/PyMuPDF. Deterministically extracts 10+ fields	services/ ocr_service.py
MatchingService	Compares OCR-extracted fields to citizen-declared values. Computes per-field and overall match scores	services/ matching_service.py
ReadinessEngine	Computes weighted 0-100 readiness score across 5 components	services/ readiness_engine.py
Rules Engine	Loads YAML service specs. Validates fields, checks eligibility, calculates fees	rules_engine/ engine.py
EventBus	In-memory async pub/sub for SSE. Routes events to application-specific subscriber queues	api/routes/ stream.py
FieldEncryptor	AES-256-GCM field-level encryption for RESTRICTED and QUASI_IDENTIFIER data at rest	data_layer/ encryption.py
RAGService	Keyword-based retrieval over markdown knowledge base. Top-5 chunks passed to LLM	services/ rag_service.py
PaymentService	Manages payment lifecycle, supports MOCK_AUTO and UPI_QR modes, OCR-verifies receipts	services/ payment_service.py
CertificateService	Generates PDF certificates for approved and paid applications	services/ certificate_service.py

11. LOW-LEVEL ARCHITECTURE

11.1 Detailed Module Map



	app/services/
PyMuPDF OCR	ocr_service.py – Tesseract +
scoring	matching_service.py – Field match
	readiness_engine.py – 5-component scorer
retrieval	rag_service.py – Keyword RAG
	payment_service.py – Payment lifecycle
	certificate_service.py – PDF generation
resolution	citizen_resolver.py – Cross-channel
	next_question_engine.py – Slot selection
strings	i18n.py – Multilingual
	language_service.py – Language detection

	app/rules_engine/
	engine.py
cache	ServiceSpecLoader – YAML → ServiceSpec
logic	FieldValidator – Field validation
evaluator	EligibilityChecker – Rule expression
evaluator	FeeCalculator – Waiver condition

	app/data_guard/
	guard.py
scanner	DataClassifier – Recursive field
writer	DataGuard – Check + audit log

	app/llm/
--	----------

	provider_factory.py	– Singleton factory
	base.py	– LLMProvider ABC
	openrouter_provider.py	
	gemini_provider.py	
	groq_provider.py	
	llm_service.py	– Unified facade
LLMUnavailableError	exceptions.py	–
	app/data_layer/	
FieldEncryptor	encryption.py	– AES-256-GCM
	repositories/	
+ FSM	application_repo.py	(40 KB) – core CRUD
	citizen_repo.py	– citizen registry
	session_repo.py	– context vault
storage	document_repo.py	– document + OCR
log	audit_repo.py	– append-only audit
identities	channel_identity_repo.py	– hashed
storage	event_repo.py	– lifecycle event
	app/models/	
	db_models.py	(499 lines)
	– All 15 SQLAlchemy model classes	

12. FRONTEND ARCHITECTURE

Source: `frontend/src/`

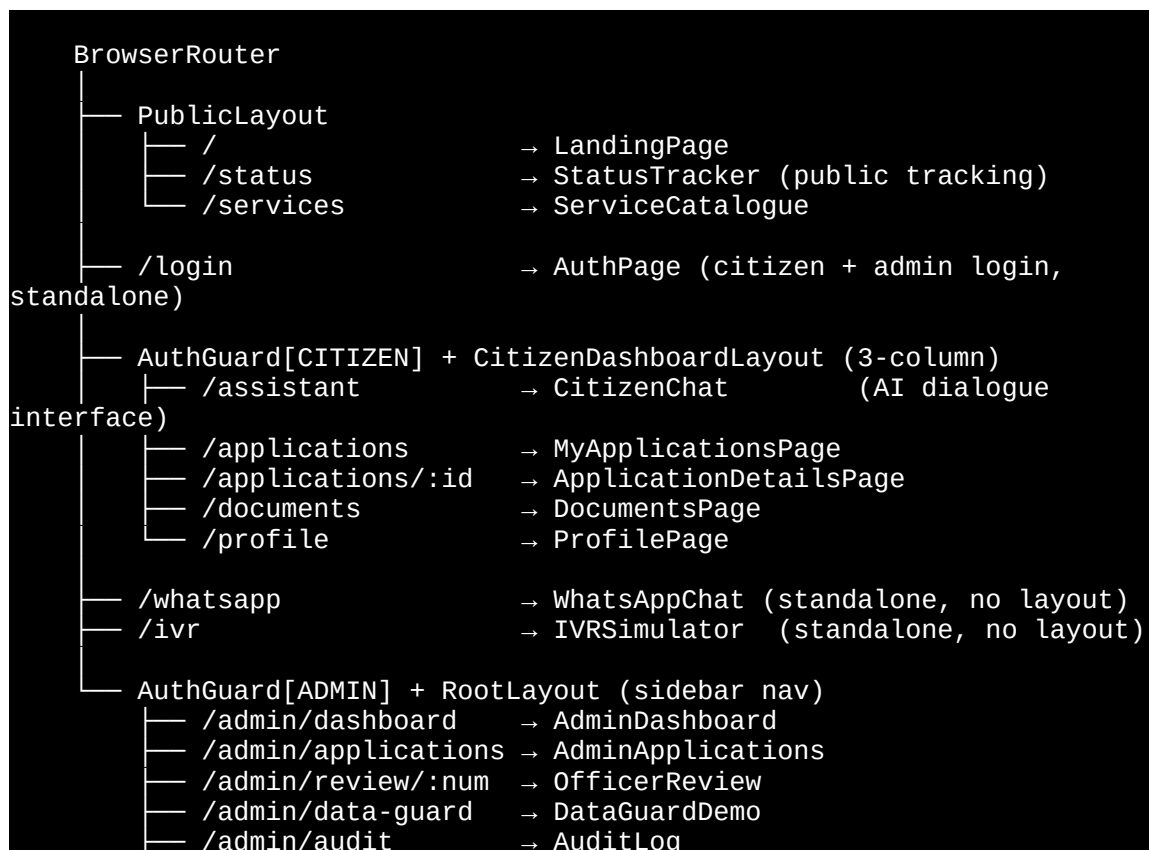
12.1 Application Shell

The frontend is a **React 19 Single Page Application** built with Vite 6.4. The entry point is `frontend/src/main.jsx`, which mounts the React root with:

- `QueryClientProvider` (react-query for server state caching)
- `BrowserRouter` (react-router-dom v7 for SPA navigation)
- `Toaster` (react-hot-toast for notification toasts)

12.2 Routing Architecture

Source: `frontend/src/App.jsx`



└─ /admin/escalations → EscalationPanel

12.3 State Management (Zustand Stores)

authStore.js — Persisted to `localStorage` under key `rsp-auth`:

```
// State shape (verified from source)
{
  token: null, // JWT for admin user
  user: null, // Admin user object { username,
role }
  isAuthenticated: false,

  citizenToken: null, // JWT for citizen user
  citizenUser: null, // { citizen_id, name, email,
phone, address }
  isCitizenAuthenticated: false,
}

// Actions:
setAuth(token, user) // Admin login
clearAuth() // Admin logout
setCitizenAuth(token, user) // Citizen login → purges old
session state
clearCitizenAuth() // Citizen logout
updateCitizenUser(fields) // Update profile fields
isAdmin() // user.role === 'ADMIN'
isCitizen() // isCitizenAuthenticated || role
=== 'CITIZEN'
```

chatStore.js — Conversational state:

- Current messages array
- Session ID and application context
- Active service type

uiStore.js — UI state:

- Sidebar open/closed
- Global loading states

12.4 API Communication Pattern

All API calls are made via `axios` with base URL

`VITE_API_URL=http://localhost:8000/api/v1`. JWT tokens are injected as `Authorization: Bearer <token>` headers. Server-Sent Events (SSE) use the `EventSource` API with `?token=<jwt>` query param (since `EventSource` does not support custom headers).

12.5 Frontend Component Architecture Diagram



13. BACKEND ARCHITECTURE

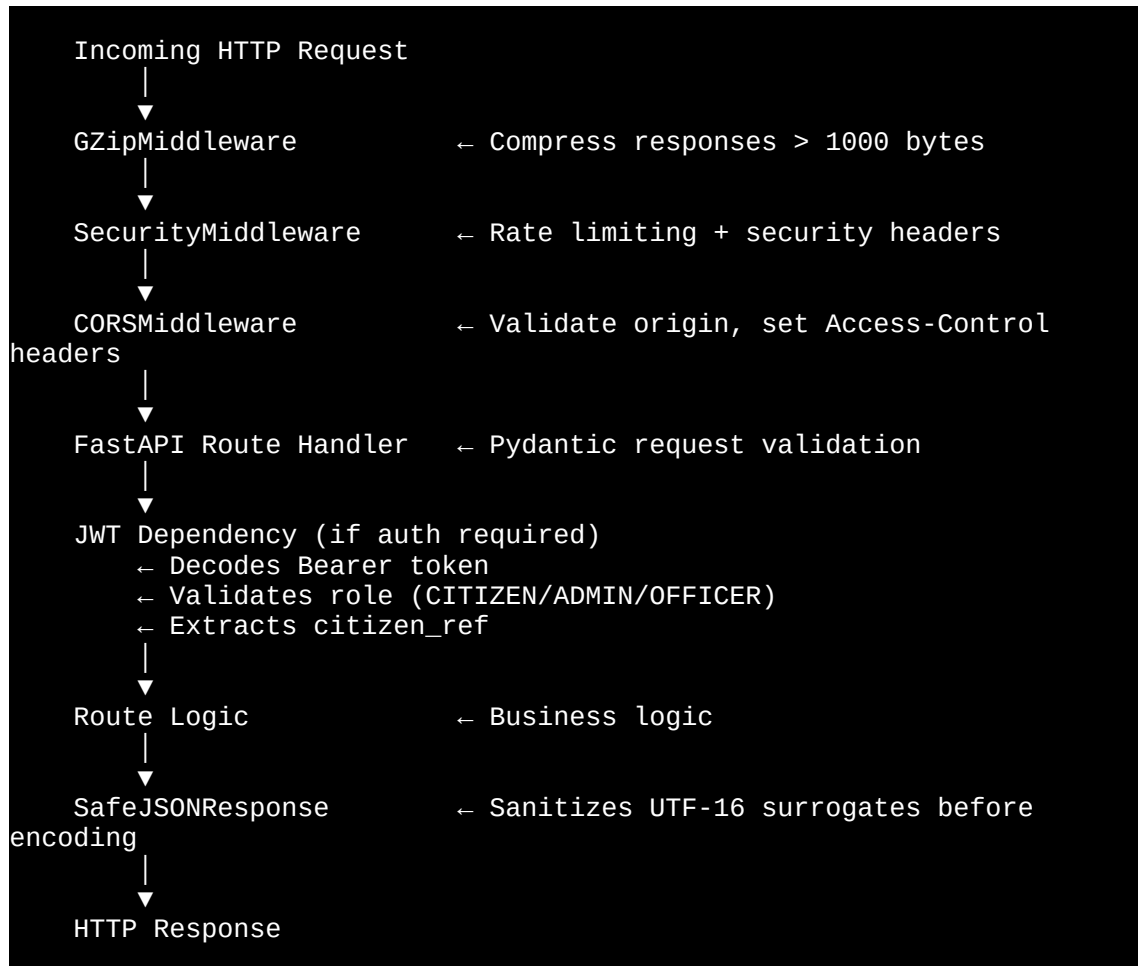
13.1 Application Startup Sequence

Source: `backend/main.py` `lifespan()` function:

```
Server Start (uvicorn)
  ↓
1. Base.metadata.create_all(bind=engine)
   → Creates all 15 SQLite tables if not present
   ↓
2. ServiceSpecLoader.load_all()
   → Reads all .yaml files from backend/seed/service_specs/
   → Parses into ServiceSpec objects
   → Caches in ServiceSpecLoader._cache dict
   ↓
3. os.makedirs() for all storage paths:
   data/uploads, data/receipts, data/certificates,
   data/audio, data/audio/ivr, data/audio/whatsapp,
   data/uploads, data/ocr_cache
   ↓
4. OCRService() initialisation
   → Auto-detects Tesseract binary from 9 candidate paths
   → Sets pytesseract.tesseract_cmd
   → Queries available languages
   ↓
5. _seed_database()
   → Creates admin user (if not exists) from
   ADMIN_USERNAME/ADMIN_PASSWORD
   → Seeds Service catalogue entries from YAML specs
   ↓
Server Ready → "Platform is ready!" logged
```

13.2 Request Processing Pipeline

Every API request goes through this middleware stack (source: `main.py`):



13.3 Conversation Processing Flow

Source: `api/routes/conversation.py` + `orchestration/state_machine/orchestrator.py`:



```

▼
ConversationOrchestrator.process_message(
    citizen_ref, text, channel, language, modality
)
    |
    |— session_repo.get_or_recover_session(citizen_ref)
    |   → Loads or creates ConversationSession
    |
    |— NLU Intent Classification
    |   → Detects: ANSWER | GREETING | HELP | QUESTION |
CHANNEL_SWITCH
    |
    |— If ANSWER intent:
    |   FieldValidator.validate_slot(slot_spec, value)
    |   → If valid: session.filled_slots[field] = value
    |   → Save to ApplicationData (encrypted if RESTRICTED)
    |
    |— NextQuestionEngine.get_next_question(session, spec)
    |   → Returns next required field to collect
    |
    |— DataGuard.check(context_payload)
    |   → Blocks if RESTRICTED fields present
    |
    |— If ALLOWED: LLMService.generate_response(context)
    |   → Returns conversational reply
    |
    |— Return {response, current_node, filled_slots,
missing_slots, ...}

```

14. DATABASE ARCHITECTURE

14.1 Overview

The database is **SQLite**, stored at `backend/revenue_services.db`. All 15 tables are defined as SQLAlchemy 2.0 ORM models in `backend/app/models/db_models.py` (499 lines).

Design Principles (verified from source code comments):

1. No raw PII stored in the `applications` table — only tokenized `citizen_ref`
2. All channel identifiers (phone, WhatsApp, email) stored as SHA-256 hashes
3. All RESTRICTED field values are AES-256-GCM encrypted in `application_data.field_value_encrypted`
4. The `audit_log` table is append-only with hash chaining for tamper detection

14.2 Entity Relationship Diagram

```
citizens (citizen_ref: "CIT-001")
  id (UUID PK)
  citizen_ref (VARCHAR UNIQUE) ← tokenized identifier
  name_encrypted, phone_encrypted, email_encrypted
  preferred_language, preferred_channel

  < channel_identities
    id (UUID PK)
    citizen_ref → FK → citizens.citizen_ref
    channel: WHATSAPP | IVR | EMAIL | WEB
    identifier_hash (SHA-256, UNIQUE per channel)
    verified (BOOL)
    UNIQUE(channel, identifier_hash)

  < users
    id (UUID PK)
    username (VARCHAR UNIQUE)
    hashed_password (sha256_crypt)
    role: CITIZEN | OFFICER | ADMIN | AUDITOR
    citizen_ref → FK → citizens.citizen_ref (nullable)

  < conversation_sessions
    id (UUID PK)
    citizen_ref → FK → citizens.citizen_ref
    application_id (UUID, nullable)
    current_node (VARCHAR) ← current FSM dialogue node
    channel: WEB | WHATSAPP | IVR | MOBILE
    language: en | hi | mr
    filled_slots (JSON) ← all collected field values
    missing_slots (JSON) ← fields still needed
    consent_given (BOOL)
    payment_status: PENDING | PAID
    anomaly_score (FLOAT)
    channel_history (JSON)
    expires_at (DATETIME)

  < applications
    id (UUID PK)
    application_number (VARCHAR UNIQUE) ← e.g. APP-0000001
    tracking_id (VARCHAR UNIQUE) ← e.g. INC-2026-000001
    citizen_ref → FK → citizens.citizen_ref
    service_id → FK → services.id
    status (VARCHAR) ← 21 FSM states
    progress_percent (INTEGER 0-100)
    payment_status: PENDING | PAID | SUCCESS
    channel_origin: WEB | WHATSAPP | IVR | MOBILE
    last_channel
    language: en | hi | mr
    consent_given (BOOL)
    anomaly_score (FLOAT)
    submitted_at, completed_at, created_at, updated_at
    < application_data
```

NON_SENSITIVE	id (UUID PK) application_id → FK → applications.id field_name (VARCHAR) field_value_encrypted (TEXT) ← AES-256-GCM classification: RESTRICTED QUASI_IDENTIFIER
WHATSAPP_TEXT	source: WEB OCR OFFICER SYSTEM
CASTE_PROOF	version (INTEGER) ← incremented on each update is_verified (BOOL)
	< documents id (UUID PK) application_id → FK → applications.id doc_type: IDENTITY_PROOF INCOME_PROOF
matching	... file_ref (VARCHAR) ← local file path upload_channel: WEB WHATSAPP IVR MOBILE raw_ocr_text (TEXT) raw_extracted_fields (JSON) ← pre-normalisation normalized_fields (JSON) ← post-normalisation extracted_fields (JSON) ← final fields used for
	field_match_scores (JSON) ← per-field match scores overall_match_score (FLOAT 0-100) mismatch_fields (JSON) ← list of mismatched field
names	
VERIFIED	verification_status: PENDING MATCHED MISMATCH
DOCUMENT_UPLOADED	confidence_score (FLOAT) normalization_status: DETERMINISTIC AI_NORMALIZED normalization_provider: LOCAL OPENROUTER
	< application_events id (UUID PK) application_id → FK → applications.id event_type (VARCHAR) ← STATUS_CHANGE
SIMULATED	... payload (JSON) sent (BOOL) ← for SSE delivery tracking created_at (DATETIME)
	< payments id (UUID PK) application_id → FK → applications.id citizen_ref → FK → citizens.citizen_ref amount (FLOAT) currency (VARCHAR, default "INR") method: UPI CARD NETBANKING CHALLAN
	status: INITIATED SUCCESS FAILED REFUNDED transaction_id (VARCHAR UNIQUE) channel: WEB WHATSAPP IVR receipt_file_ref (VARCHAR) created_at, completed_at
	< escalations id (UUID PK) application_id → FK → applications.id raised_by, assigned_to, reason, resolution status: OPEN ASSIGNED RESOLVED CLOSED

```

└─1 certificate
    id (UUID PK)
    application_id → FK → applications.id (UNIQUE)
    citizen_ref
    certificate_number (VARCHAR UNIQUE)
    file_ref (VARCHAR) ← local PDF path
    issue_date (DATETIME)
    digital_signature (TEXT)

services (seeded from YAML on startup)
| id (VARCHAR 64 PK) ← "income_certificate"
| name_en, name_hi, name_ta, name_te
| department (VARCHAR)
| fee_amount (FLOAT), fee_currency
| sla_days (INTEGER)
| required_docs (JSON)
| waiver_conditions (JSON)

audit_logs (INTEGER PK, APPEND-ONLY)
| id (Integer auto-increment)
| event_type: DATA_GUARD | CONSENT | PAYMENT | SUBMISSION | AUTH
| STATUS_UPDATE
| actor (VARCHAR) ← username or "CITIZEN" or "SYSTEM"
| citizen_ref, application_id
| action (TEXT) ← human-readable description
| outcome: ALLOW | BLOCK | SUCCESS | FAILURE
| blocked_fields (JSON)
| metadata (JSON)
| payload_hash (VARCHAR) ← SHA-256 of payload
| previous_hash (VARCHAR) ← links to prior audit entry
| created_at (DATETIME)

ivr_sessions
| id (UUID PK)
| caller_phone_hash (VARCHAR) ← SHA-256 of phone number
| citizen_ref, application_id
| call_state: GREETING | LANGUAGE_SELECT | IDENTIFY | MAIN_MENU |
...
| call_id (VARCHAR)
| created_at, updated_at

conversation_messages
| id (UUID PK)
| session_id → FK → conversation_sessions.id
| role: CITIZEN | ASSISTANT | SYSTEM
| content (TEXT)
| language: en | hi | mr
| modality: TEXT | VOICE | DTMF
| created_at (DATETIME)

```

14.3 Important Table Notes

applications table: Contains NO raw PII. The `citizen_ref` column is a tokenized ID (e.g. "CIT-001"), not the citizen's actual name or phone number. The `service_id` is a string key like "income_certificate". The `status` column holds the current FSM state as a string.

application_data table: Each individual form field is stored as a separate row with its encryption, classification, and data source tracked independently. The `version` column increments on each update, supporting field-level provenance tracking.

documents table: Stores both the raw OCR output (`raw_ocr_text`, `raw_extracted_fields`) and the normalised output (`normalized_fields`, `extracted_fields`) separately. This allows comparing the original OCR noise vs. the cleaned values. The `field_match_scores` JSON maps field names to their individual match scores (0-100), while `overall_match_score` is the aggregate.

audit_logs table: The `previous_hash` column contains the SHA-256 hash of the preceding audit log entry's key fields. This creates a blockchain-style chain where any tampering with a past entry breaks the chain from that point forward.

15. AI / LLM ARCHITECTURE

15.1 Provider Factory Pattern

Source: `backend/app/llm/provider_factory.py`

The LLM provider is instantiated **once per server lifetime** using a singleton pattern:

```
_instance: Optional[LLMProvider] = None

def get_provider() -> LLMProvider:
    global _instance
    if _instance is not None:
```

```

        return _instance

    p = settings.LLM_PROVIDER.lower()
    if p == "openrouter":
        _instance = OpenRouterProvider()
    elif p == "gemini":
        _instance = GeminiProvider()
    elif p == "groq":
        _instance = GroqProvider()
    else:
        raise LLMUnavailableError(f"Unknown LLM provider: {p}")

    return _instance

```

Critical policy: There is **NO fallback** to another provider or to local keyword matching. If the configured provider fails to initialise (missing API key, wrong key format), `LLMUnavailableError` is raised immediately. The FastAPI global exception handler returns HTTP 503 with a user-friendly message. The server does not silently degrade.

15.2 Provider Configurations

Provider	Config Variable	Default Model	Notes
OpenRouter	<code>OPENROUTER_API_KEY</code>	<code>minimax/minimax-m3:free</code>	200+ models available. Supports free tier. HTTP POST to <code>https://openrouter.ai/api/v1/chat/completions</code>
Gemini	<code>GEMINI_API_KEY</code>	<code>gemini-1.5-flash</code>	Google AI Studio. Uses <code>google-generativeai</code> SDK
Groq	<code>GROQ_API_KEY</code>	<code>llama3-8b-8192</code>	Extremely low latency inference. Compatible with OpenAI API format

Startup validation (source: `app/core/config.py validate_llm_provider()`): The server refuses to start if the configured provider's API key:

- Is `None` or empty
- Contains placeholder strings like `"your_key"`, `"replace_with_real_key"`, `"sk-or-v1-"` without a real key

15.3 LLM Usage Contexts

The LLM is called in exactly three contexts, all of which pass through Data Guard:

```
Context 1: Dialogue Generation (ConversationOrchestrator)
  Input: Application context (service_id, current_node,
missing_slots)
  Guard: DataGuard.check() → NO citizen PII in payload
  Output: Conversational reply to citizen

Context 2: OCR Field Normalization (OCRService)
  Input: Cleaned OCR text (max 1500 chars) + raw extracted fields
  Guard: DataGuard.check() with data_classification="SYNTHETIC"
  Output: Normalized field values + corrections list
  Fallback: If Guard blocks or LLM unavailable → use deterministic
normalization

Context 3: RAG Answer Generation (LLMService)
  Input: Top-5 retrieved knowledge chunks (no citizen data)
  Guard: Already filtered – chunks are static .md file text
  Output: Citizen-friendly explanation of eligibility/process
```

15.4 AI/LLM Architecture Diagram

```
graph LR
    CM["Citizen Message"]

    subgraph NLU ["NLU Layer"]
        IC["Intent Classifier<br/>(ANSWER/HELP/QUESTION)"]
        EE["Entity Extractor<br/>(field values)"]
    end

    subgraph ORCH ["Orchestrator"]
        FV["Field Validator<br/>(deterministic)"]
        NQE["Next Question<br/>Engine"]
    end

    subgraph DG_BLOCK ["🛡️ Data Guard Firewall"]
        DC["DataClassifier<br/>scan_payload()"]
        ALLOW["✅ ALLOWED<br/>(no PII detected)"]
        BLOCK["🚫 BLOCKED<br/>(PII detected)"]
    end

    subgraph AI_LAYER ["AI Layer (External)"]
        RAG["RAGService<br/>(keyword retrieval)"]
        KB["Knowledge Base<br/>(.md files)"]
        LLM["LLM Provider<br/>(OpenRouter/Gemini/Groq)"]
    end
```

```
DETERM["Deterministic<br/>Fallback Response"]
```

```
CM --> IC & EE
IC & EE --> FV --> NQE
NQE --> DC
DC --> ALLOW & BLOCK
ALLOW --> RAG
RAG --> KB
KB --> RAG --> LLM
LLM --> RESP["Response to Citizen"]
BLOCK --> DETERM --> RESP
```

```
style BLOCK fill:#ff6b6b,color:#fff
style ALLOW fill:#51cf66,color:#fff
style LLM fill:#4dabf7,color:#fff
```

16. OCR ARCHITECTURE

16.1 OCR Pipeline Overview

Source: `backend/app/services/ocr_service.py` (702 lines)

The OCR pipeline is **100% deterministic** at its core. The optional AI normalisation step is an enhancement that can be gracefully skipped if Data Guard blocks it or the LLM is unavailable.

```
Document File (image or PDF)
|
▼
Step 1: _extract_text(file_path, language)
|   |   |   |   |
|   |   |   |   └─ If PDF:
|   |   |   |       └─ Try PyMuPDF (fitz): page.get_text() for all pages
|   |   |   |       └─ Try pypdf: PdfReader + extract_text()
|   |   |   |       └─ If above fail: Render pages to images → Tesseract OCR
|   |   |   └─ If Image:
|   |   |       └─ pytesseract.image_to_string(img, lang, config="--oem 3
--psm 6")
|   |   └─ Fallback: _mock_extract() (deterministic test mock)
|   ▼
Step 2: _clean_ocr_text(raw_text)
|   |   |
|   |   └─ Normalize line breaks (\r\n → \n)
|   |   └─ Strip OCR artifacts: ~ | ^ _ « » < > { } [ ] ;
```

```

└─ Collapse multiple spaces
↓
Step 3: detect_document_type(cleaned_text)
├─ Keyword scoring: DOC_TYPE_KEYWORDS dict (multilingual
en/hi/mr)
├─ Returns: AADHAAR_CARD | PAN_CARD | VOTER_ID | INCOME_PROOF
|
|          SALARY_SLIP | CASTE_CERTIFICATE |
DOMICILE_CERTIFICATE |
|          BANK_PASSBOOK | PAYMENT_RECEIPT | DRIVING_LICENSE
| UNKNOWN
↓
Step 4: _extract_fields(cleaned_text, doc_type)
├─ 1. Aadhaar: \b(\d{4}\s\d{4}\s\d{4}|\d{12})\b
├─ 2. DOB: labeled pattern + standalone DD/MM/YYYY fallback
├─ 3. Gender: labeled keyword (male/female/पुरुष/महिला)
├─ 4. PAN: [A-Z]{5}[0-9]{4}[A-Z] exact pattern
├─ 5. Name: labeled "name:" pattern + heuristic (2-4 alpha
words)
├─ 6. Income: ₹/Rs./INR labeled match for annual/gross/net
income
├─ 7. Employer: company/organisation label
├─ 8. Address: address/पत्ता label
├─ 9. Transaction ID: txn/reference/upi-ref label
└─ 10. Caste: SC|ST|OBC|OPEN|GENERAL keyword
↓
Step 5: _normalize_deterministic_fields(raw_fields)
├─ Names: strip stop-words (shri, mr, mrs, श्री), OCR artifacts
├─ PAN: uppercase, validate 10-char [A-Z]{5}[0-9]{4}[A-Z]
├─ DOB: normalize to DD-MM-YYYY (handles YYYY-MM-DD, DD/MM/YY)
├─ Income: strip ₹ commas → integer string
├─ Aadhaar: strip spaces → exactly 12 digits
└─ Gender: normalize to "Male" | "Female" | "Transgender"
↓
Step 6: [Optional] DataGuard.check() + LLM.normalize_ocr_fields()
├─ Check payload with data_classification="SYNTHETIC"
├─ If ALLOWED: OpenRouter normalises ambiguous fields
├─ Merges AI corrections into normalized_fields dict
└─ If BLOCKED or error: use deterministic fields (no crash)
↓
OCRResult(
    raw_text,          ← original Tesseract output
    raw_fields,        ← pre-normalisation extracted fields
    normalized_fields, ← post-normalisation fields
    extracted_fields,  ← final fields for matching
    doc_type_detected,
    confidence,        ← 0.90 (Tesseract) | 0.75 (PyMuPDF) | 0.0
(mock)
    provider,          ← "tesseract" | "pymupdf_or_mock" | "mock"
    normalization_metadata: { status, provider, ai_normalized,
corrections }
)

```

16.2 Tesseract Auto-Detection Priority

The `_init_tesseract()` method checks the following paths in order:

1. `settings.TESSERACT_CMD` (from `.env`)
2. `settings.TESSERACT_PATH` (from `.env`)
3. `os.getenv("TESSERACT_CMD")` (system env var)
4. `os.getenv("TESSERACT_PATH")` (system env var)
5. `shutil.which("tesseract")` (system PATH)
6. `C:\Program Files\Tesseract-OCR\tesseract.exe` (Windows default)
7. `C:\Program Files (x86)\Tesseract-OCR\tesseract.exe` (Windows x86)
8. `/usr/bin/tesseract` (Linux)
9. `/usr/local/bin/tesseract` (macOS)
10. `/opt/homebrew/bin/tesseract` (macOS Homebrew)

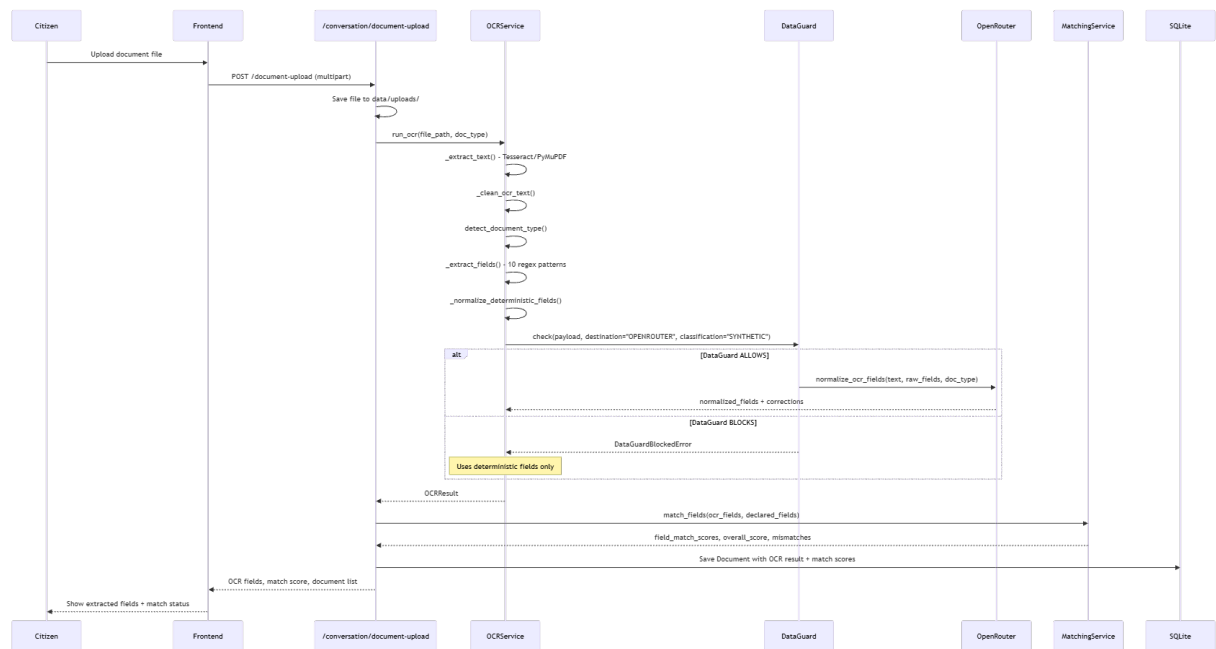
16.3 OCR Normalization: Example

The `_normalize_name()` method (source: lines 299–335) handles noisy OCR output:

```
Raw OCR text: "Ee Ae D Viki Bhausaheb Lokhande Oa Bs Ga"
|
▼ Strip stop-words: "shri", "mr", etc.
▼ Strip 1-2 char artifacts from start/end
|
Result: "Viki Bhausaheb Lokhande"
```

The matching service then compares this normalised value against the citizen's declared `applicant_name`, producing a score of 100 if exact match.

16.4 OCR Sequence Diagram



17. DATA GUARD ARCHITECTURE

17.1 Overview

The **Data Guard** is the most critical security component of the platform. It acts as a mandatory gateway between the application's data layer and any external AI API call. It cannot be bypassed — every LLM call must pass a `DataGuard.check()` first.

Source: `backend/app/data_guard/guard.py` (262 lines)

17.2 Data Classification Schema

RESTRICTED_FIELDS — ANY of these in a payload causes immediate BLOCK:

```
RESTRICTED_FIELDS = {
    "aadhaar_number",    # National identity number
    "pan_number",        # Tax identity number
    "voter_id",          # Election identity
```

```

    "passport_number",    # Travel document
    "bank_account",      # Financial account
    "applicant_dob",     # Date of birth
    "date_of_birth",     # Alias
    "biometric_data",    # Fingerprint/iris/face
    "medical_history",   # Health records
    "applicant_name",    # Personal name
    "father_name",       # Family PII
    "mother_name",       # Family PII
    "phone_number",      # Contact PII
    "email_address",     # Contact PII
    "address",           # Location PII
    "house_number",      # Location PII
    "street",            # Location PII
    "full_name",         # Alias for applicant_name
}

```

QUASI_IDENTIFIER_FIELDS — Blocked unless payload is tagged as **SYNTHETIC** AND quasi-identifier count < 3:

```

QUASI_IDENTIFIER_FIELDS = {
    "district",
    "annual_income",
    "caste_category",
    "caste_name",
    "occupation",
    "age_range",
    "residence_years",
    "place_of_birth",
    "pincode",
}

```

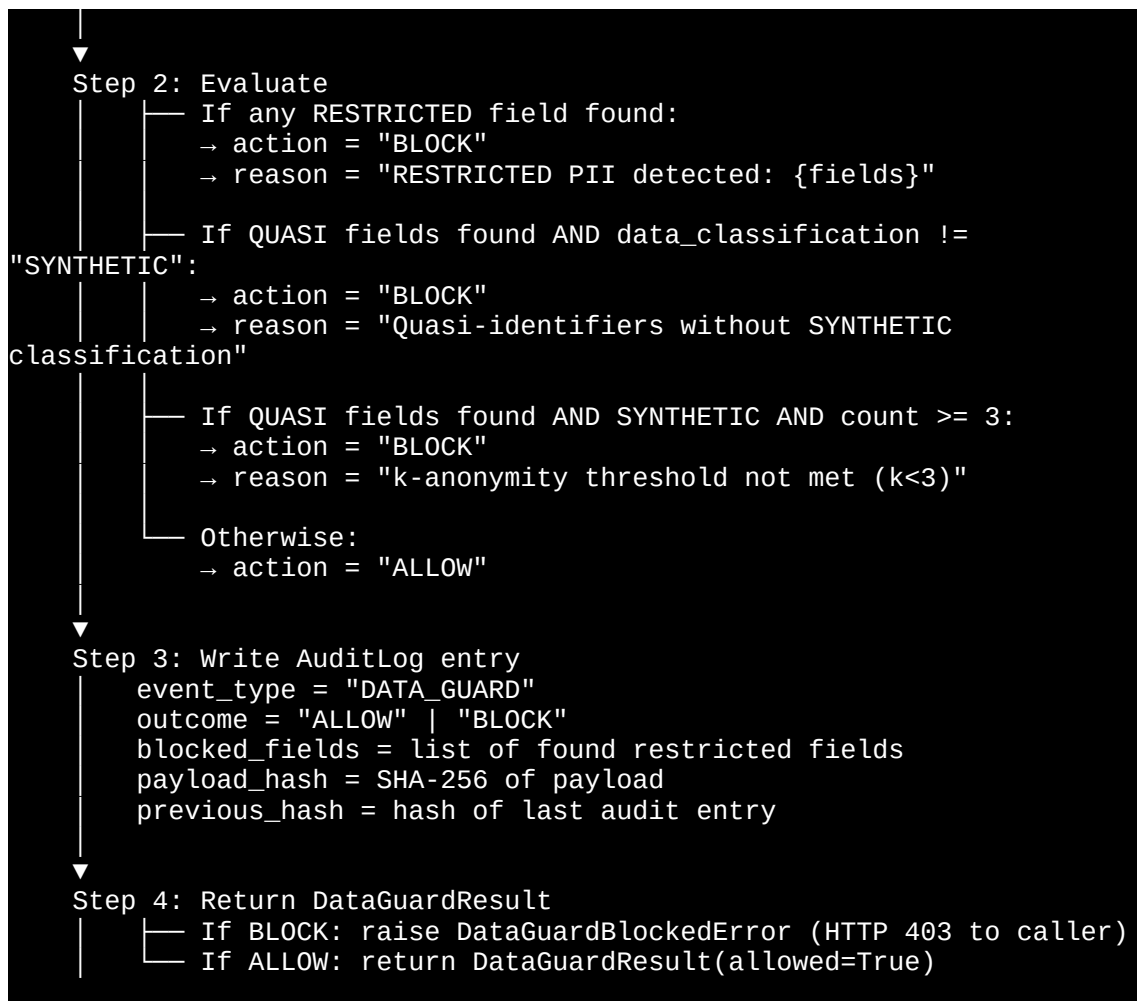
NON_SENSITIVE_FIELDS — Everything else. Always allowed.

17.3 Data Guard Decision Logic

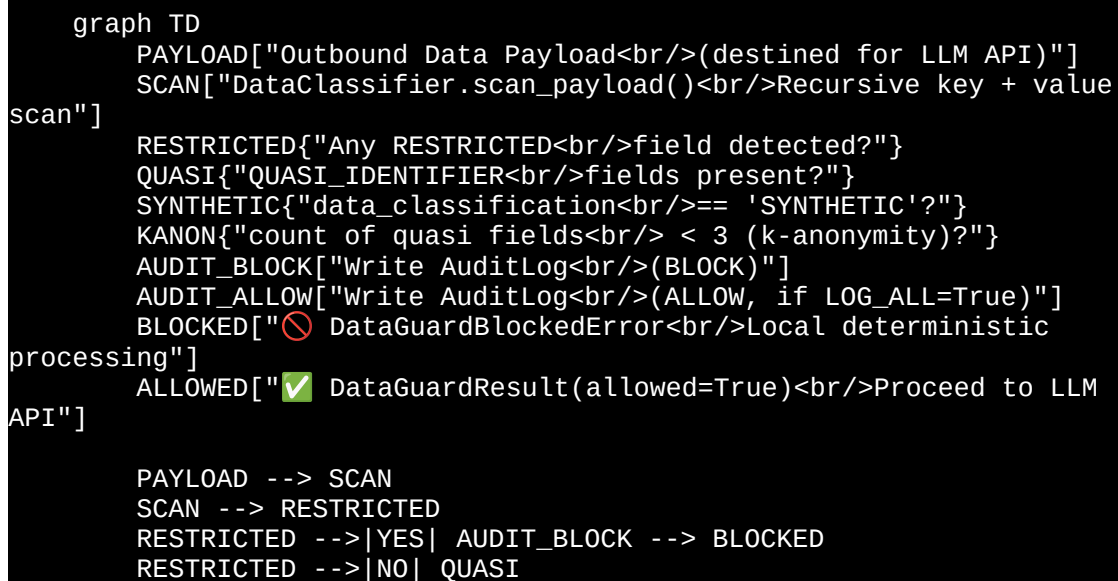
```

DataGuard.check(payload, destination, caller, operation, ...)
|
▼
Step 1: DataClassifier.scan_payload(payload)
|   |   |   |   |
|   |   |   |   |— Recursively scan all dict keys
|   |   |   |   |— Check each key against RESTRICTED_FIELDS (exact match)
|   |   |   |   |— Check text values for Aadhaar pattern: \b\d{4}\s?\d{4}\s?\d{4}\b
|   |   |   |   |— Returns: DataScanResult(restricted_fields, quasi_fields, blocked)

```



17.4 Data Guard Flow Diagram



```

QUASI -->|YES| SYNTHETIC
QUASI -->|NO| AUDIT_ALLOW --> ALLOWED
SYNTHETIC -->|NO| AUDIT_BLOCK --> BLOCKED
SYNTHETIC -->|YES| KANON
KANON -->|YES| AUDIT_ALLOW --> ALLOWED
KANON -->|NO| AUDIT_BLOCK --> BLOCKED

style BLOCKED fill:#ff6b6b,color:#fff
style ALLOWED fill:#51cf66,color:#fff

```

⚠ **CRITICAL SECURITY WARNING:** `DATA_GUARD_ENABLED` must **NEVER** be set to `False` in any environment processing real citizen data. Disabling Data Guard would allow Aadhaar numbers, PAN cards, dates of birth, and personal names to be transmitted to external commercial AI APIs in plaintext. This would constitute a serious data protection violation.

18. RAG ARCHITECTURE

18.1 Overview

Source: `backend/app/services/rag_service.py` (211 lines)

The RAG (Retrieval-Augmented Generation) service provides factual, source-verified answers to citizen questions about eligibility, required documents, and application processes. It uses a **keyword-based retrieval approach** — no vector database or embedding model is required.

18.2 Knowledge Base Structure

```

it backend/knowledge/
    ├── income_certificate/
    │   ├── overview.md      ← What is income certificate, who needs
    │   ├── documents.md     ← Required documents with explanations
    │   ├── eligibility.md   ← Eligibility criteria and income limits
    │   ├── faq.md           ← Frequently asked questions
    │   └── process.md       ← Step-by-step application process
    └── caste_certificate/   ← Same 5-file structure

```

```
└─ domicile_certificate/ ← Same 5-file structure
└─ obc_ncl_certificate/ ← Same 5-file structure
```

18.3 RAG Processing Flow

```
RAGService._load_all_chunks() ← Runs once (class-level singleton)
└─ KNOWLEDGE_DIR.rglob("*.md")
    └─ _parse_markdown() → split by H2 (##) and H3 (###)
        └─ Creates KnowledgeChunk(text, source_path, heading)
↓
RAGService.retrieve(question, service_id, max_chunks=5)
└─ question_tokens = _tokenize(question)
    └─ lowercase words, filter 30+ stopwords
    └─ For each chunk in _chunks:
        └─ score = _score_chunk(chunk, question_tokens, service_id)
            └─ overlap = |Q_tokens ∩ chunk_tokens|
            └─ base_score = overlap / |Q_tokens|
            └─ If service_id matches chunk.source: score *= 1.5
            └─ If "general" in chunk.source: score *= 1.1
        └─ Return top max_chunks by score (text strings)
↓
LLMService.answer_rag(question, chunks, language)
└─ Prompt: "Answer ONLY using these knowledge chunks:
{chunks}"
└─ LLM generates answer without hallucinating
```

19. RULES ENGINE ARCHITECTURE

19.1 Overview

Source: `backend/app/rules_engine/engine.py` (353 lines)

The Rules Engine is the **deterministic business logic backbone** of the platform. It loads service specifications from YAML files, validates form fields, checks eligibility, and calculates fees — all without involving the LLM.

19.2 ServiceSpec Loading

```
Application Startup → ServiceSpecLoader.load_all()
├── Scan settings.SERVICE_SPECS_DIR (or
backend/seed/service_specs/)
│   ├── income_certificate.yaml
│   ├── caste_certificate.yaml
│   ├── domicile_certificate.yaml
│   └── obc_ncl_certificate.yaml
│
│   For each YAML:
│   → Parse service metadata (id, name_en/hi, department, sla_days,
fee)
│   → Parse slots[] → List[SlotSpec]
│   → Parse documents.required[] and documents.optional[]
│   → Parse eligibility[] rules
│   → Parse cross_field_validations[]
│   → Cache in ServiceSpecLoader._cache[service_id]
```

19.3 Field Validation

Source: `FieldValidator.validate_slot()`:

```
Input: SlotSpec + citizen_provided_value
├── If value is None or empty:
│   └── Return (False, "field is required") if slot.required
├── If slot.type == "number":
│   ├── Parse as float
│   ├── Check  $v["min"] \leq \text{value} \leq v["max"]$ 
│   └── Return error_msg from spec if fail
├── If slot.type == "string":
│   ├── Anti-fake-name guard: if field is "applicant_name" AND
│   │   value contains "certificate", "income", "caste", "domicile"
│   │   → Return "Please enter a valid person's full name"
│   ├── Check min_length / max_length
│   ├── Check regex pattern (if defined in spec)
│   └── Check allowed_values enum (if defined)
├── If slot.type == "date":
│   └── Validate DD-MM-YYYY format: ^\d{2}-\d{2}-\d{4}$
└── Return (True, "") if all checks pass
```

19.4 Eligibility Rule Evaluation

The `EligibilityChecker` evaluates rule strings like `"annual_income <= 300000"` without using `eval()`:

```
SAFE_OPERATORS = {
    ">=": lambda a, b: a >= b,
    "<=": lambda a, b: a <= b,
    ">": lambda a, b: a > b,
    "<": lambda a, b: a < b,
    "==": lambda a, b: str(a).upper() == str(b).upper(),
    "!=": lambda a, b: str(a).upper() != str(b).upper(),
    "in": lambda a, b: str(a).upper() in [str(x).upper() for x in
b],
}
```

Special handling: If `applicant_age` appears in a rule and `applicant_dob` is in the data, age is computed dynamically as `(today - dob).days // 365`.

19.5 Fee Calculation

The `FeeCalculator` applies waiver conditions from the YAML spec:

```
# Example from income_certificate.yaml (structure verified)
fee:
  amount: 50
  currency: INR
  waiver_conditions:
    - condition: "annual_income < 100000"
      waiver_percent: 100
    - condition: "disability_status == True"
      waiver_percent: 50
```

The calculator evaluates each condition against citizen data in order. The first matching condition wins. If no condition matches, full fee applies.

20. APPLICATION LIFECYCLE — FSM

20.1 State Definitions

Source:

backend/app/orchestration/state_machine/application_fsm.py (451 lines)

The `AppState` class defines 21 states grouped into 7 phases:

```
Phase 1 – INITIATION
  INITIATED           → Application record created
  CONSENT_GIVEN       → Citizen has accepted terms
  SERVICE_SELECTED    → Service type chosen
(income/caste/domicile/obc)

Phase 2 – DATA COLLECTION
  INFORMATION_COLLECTION → Collecting form fields via AI dialogue
(self-loop)
  DOCUMENT_COLLECTION   → Uploading and verifying documents (self-loop)

Phase 3 – OCR & VALIDATION
  OCR_PROCESSING        → Running Tesseract OCR on uploaded documents
  VALIDATION_COMPLETED → OCR + matching complete, results available

Phase 4 – READINESS ASSESSMENT
  READINESS_CHECK       → Computing 5-component readiness score
  READY_FOR_REVIEW      → Score ≥ 75, no blocking issues → can submit
  FIX_REQUIRED          → Score < 75 or blocking issues → citizen must fix

Phase 5 – FINAL REVIEW
  FINAL_REVIEW          → Citizen reviews all data before submitting
  CONSENT_CONFIRMED     → Citizen confirms submission consent

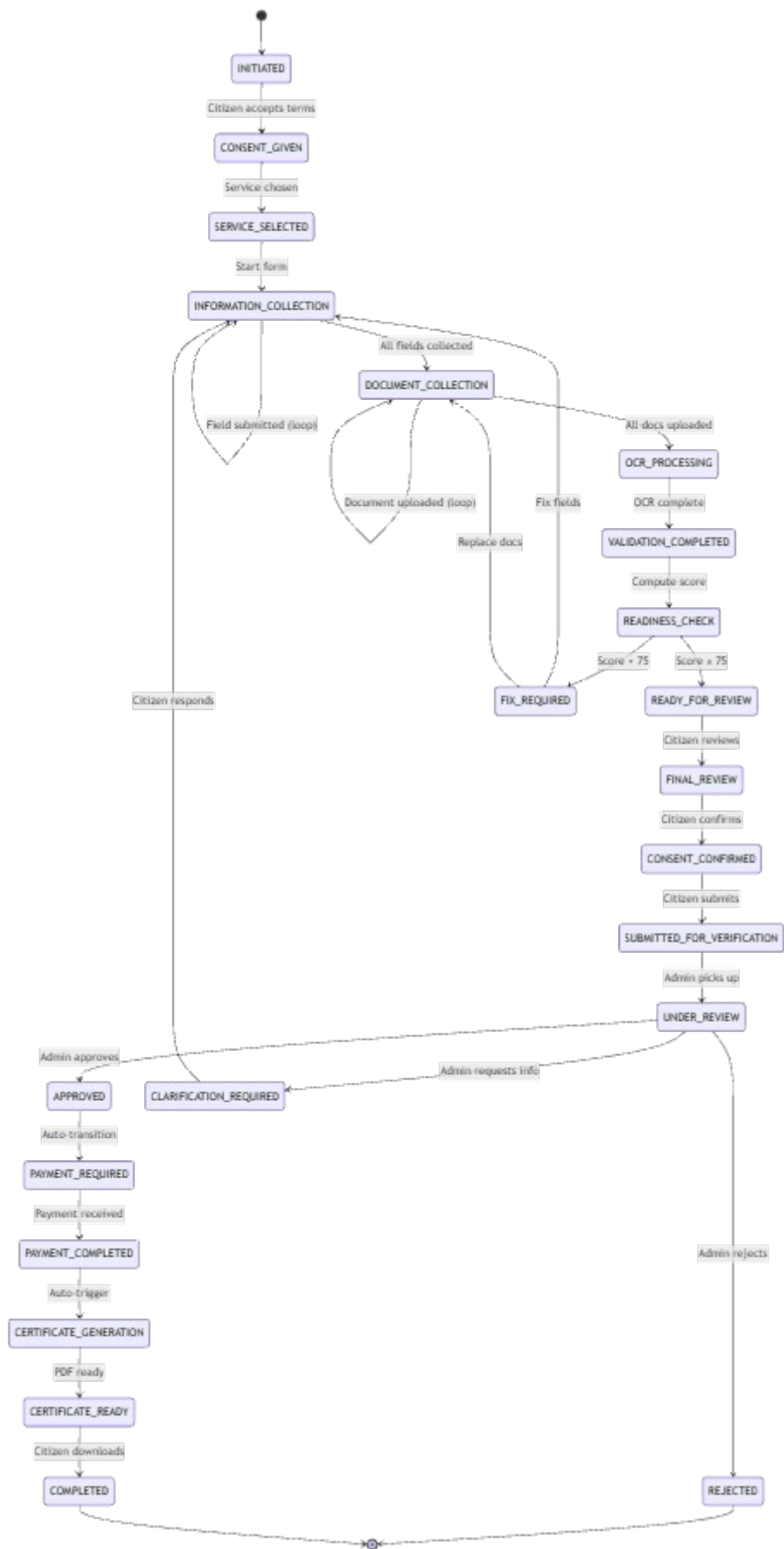
Phase 6 – GOVERNMENT REVIEW
  SUBMITTED_FOR_VERIFICATION → Sent to government review queue
  UNDER_REVIEW           → Officer is actively reviewing
  CLARIFICATION_REQUIRED → Officer requests additional information
  APPROVED               → Officer approves the application
  REJECTED               → Officer rejects (TERMINAL)
  PAYMENT_REQUIRED       → Triggered by APPROVED → citizen must pay
  PAYMENT_COMPLETED      → Payment received and verified
```

Phase 7 – CERTIFICATE
 CERTIFICATE_GENERATION → Generating PDF certificate
 CERTIFICATE_READY → PDF available for download
 COMPLETED → Process complete (TERMINAL)

20.2 State Progress Mapping

State	Progress %	UI Indicator
INITIATED	3%	Starting
CONSENT_GIVEN	8%	Consent
SERVICE_SELECTED	12%	Service
INFORMATION_COLLECTION	25%	Filling Form
DOCUMENT_COLLECTION	38%	Documents
OCR_PROCESSING	48%	Scanning
VALIDATION_COMPLETED	55%	Validated
READINESS_CHECK	60%	Checking
FIX_REQUIRED	57%	Fix Needed
READY_FOR_REVIEW	65%	Ready
FINAL_REVIEW	70%	Review
CONSENT_CONFIRMED	73%	Confirmed
SUBMITTED_FOR_VERIFICATION	78%	Submitted
UNDER_REVIEW	83%	Under Review
CLARIFICATION_REQUIRED	80%	Clarification
APPROVED	88%	Approved 
REJECTED	100%	Rejected 
PAYMENT_REQUIRED	90%	Pay Now
PAYMENT_COMPLETED	93%	Paid
CERTIFICATE_GENERATION	96%	Generating
CERTIFICATE_READY	99%	Ready
COMPLETED	100%	Done 

20.3 FSM State Transition Diagram



20.4 Locked States

In the following states, citizens **cannot modify** their application data. Any attempt to update fields while in a locked state is rejected:

```
LOCKED_STATES = {
    SUBMITTED_FOR_VERIFICATION, UNDER_REVIEW,
    CLARIFICATION_REQUIRED,
    APPROVED, PAYMENT_REQUIRED, PAYMENT_COMPLETED,
    CERTIFICATE_GENERATION, CERTIFICATE_READY, COMPLETED, REJECTED,
}
```

20.5 FSM Class Interface

```
class ApplicationFSM:

    def __init__(self, current_state: str = AppState.INITIATED)

    @property
    def state(self) -> str          # Current state

    @property
    def progress(self) -> int       #
STATE_PROGRESS[current_state]

    @property
    def is_terminal(self) -> bool   # REJECTED or COMPLETED

    @property
    def is_locked(self) -> bool     # In LOCKED_STATES

    def can_transition_to(self, new_state: str) -> bool
    # Returns True if transition is in
VALID_TRANSITIONS[current_state]

    def transition(self, new_state: str) -> Tuple[bool, str]
    # Returns (success, label_or_error_message)

    def get_next_states(self) -> List[str]
    # Returns all valid states reachable from current state

    def get_citizen_message(self, language: str = "en") -> str
    # Returns pre-written citizen message in en/hi/mr

    def to_dict(self) -> dict
    # Serializes current state, progress, transitions, messages
```

20.6 Multilingual Citizen Messages

Every FSM state has a pre-written citizen-facing message in English, Hindi, and Marathi.
Example for `PAYMENT_REQUIRED`:

- **en:** "✅ Your application is approved! Please complete the ₹50 payment to receive your certificate."
 - **hi:** "✅ आपका आवेदन स्वीकृत हुआ! प्रमाण पत्र प्राप्त करने के लिए ₹50 का भुगतान करें।"
 - **mr:** "✅ तुमचा अर्ज मंजूर झाला! प्रमाणपत्र मिळवण्यासाठी ₹50 भरा."
-

21. AUTHENTICATION & AUTHORIZATION ARCHITECTURE

21.1 Authentication Flows

Admin/Officer Login:

```
POST /api/v1/auth/login
├── Form data: username, password
├── Query User table by username
├── passlib.CryptContext.verify(password, user.hashed_password)
│   Algorithm: sha256_crypt
├── Create JWT payload: { sub, role, exp, iat }
└── Return: { access_token, token_type: "bearer" }
```

Citizen Registration:

```
POST /api/v1/auth/citizen/register
├── Body: { name, phone, email, address, preferred_language }
├── Check for existing citizen by phone or email
├── Create Citizen record with citizen_ref "CIT-{padded_id}"
├── Create User record with hashed_password (sha256_crypt)
├── Create ChannelIdentity records:
│   ├── WHATSAPP: identifier_hash = SHA-256(phone)
│   ├── IVR: identifier_hash = SHA-256(phone)
│   └── EMAIL: identifier_hash = SHA-256(email)
└── Return JWT: { citizen_ref, name, ... }
```

Citizen Login:

```
POST /api/v1/auth/citizen/login
├─ Body: { identifier (phone or email), password }
├─ Resolve citizen from identifier
├─ Verify password
├─ Create JWT payload: { sub, role: "CITIZEN", citizen_ref, exp }
└─ Return: { access_token, citizen_id, name, ... }
```

21.2 JWT Token Structure

```
{
  "sub": "username_or_citizen_id",
  "role": "CITIZEN | ADMIN | OFFICER | AUDITOR",
  "citizen_ref": "CIT-001",
  "exp": 1750000000,
  "iat": 1749971200
}
```

Token Settings:

- Algorithm: HS256
- Secret: settings.SECRET_KEY
- Expiry: 480 minutes (8 hours)
- Library: python-jose[cryptography]

21.3 Route Protection Dependencies

```
# Defined in app/api/routes/auth.py and app/core/security.py

def get_current_user(token: str = Depends(oauth2_scheme), db = Depends(get_db)):
    # Decodes JWT, looks up User record
    # Raises 401 if expired or invalid

def get_current_citizen(token = Depends(get_current_user)):
    # Verifies role == "CITIZEN"
    # Raises 403 if not a citizen

def require_admin(token = Depends(get_current_user)):
    # Verifies role in ("ADMIN", "OFFICER")
```

```
# Raises 403 if not admin/officer
```

21.4 Citizen Data Isolation

Every application query enforces `citizen_ref` ownership:

```
# From api/routes/applications.py
@router.get("/my-applications")
def get_my_applications(current_citizen =
Depends(get_current_citizen), ...):
    apps = repo.get_by_citizen(current_citizen.citizen_ref,
limit=50)
    # NEVER returns another citizen's applications

# From api/routes/applications.py
if target_citizen.citizen_ref != current_citizen.citizen_ref:
    raise HTTPException(status_code=403, detail="Forbidden: You
cannot access another citizen's applications.")
```

The SSE stream also enforces isolation: a citizen's SSE subscription is keyed to their `citizen_ref`, so only events destined for that citizen's applications are delivered.

21.5 Security Architecture Diagram

```
graph TD
    C[" Citizen Request"]
    A[" Admin Request"]

    subgraph MW["Middleware Stack"]
        RATE["Rate Limiter<br/>(SecurityMiddleware)"]
        CORS["CORS Validation"]
        JWT["JWT Decoder<br/>python-jose HS256"]
    end

    subgraph AUTHZ["Authorization Layer"]
        CITIZEN_GUARD["get_current_citizen()<br/>role == CITIZEN"]
        ADMIN_GUARD["require_admin()<br/>role in ADMIN,OFFICER"]
        OWNERSHIP["verify_application_ownership()<br/>citizen_ref
match"]
    end

    subgraph DATA["Data Security"]
        ENC["AES-256-GCM<br/>FieldEncryptor"]
        HASH["SHA-256<br/>Channel Identity Hash"]
    end
```

```

        DG["DataGuard<br/>PII Firewall"]
    end

    DB["SQLite"]

    C --> RATE --> CORS --> JWT
    A --> RATE --> CORS --> JWT
    JWT --> CITIZEN_GUARD & ADMIN_GUARD
    CITIZEN_GUARD --> OWNERSHIP
    OWNERSHIP --> ENC --> DB
    ADMIN_GUARD --> DB
    ENC --> HASH
    DG -->|"blocks PII"| JWT

    style DG fill:#ff6b6b,color:#fff
    style ENC fill:#fcc419,color:#000

```

22. API ARCHITECTURE & REFERENCE

All endpoints are prefixed with `/api/v1` (configured via `settings.API_V1_PREFIX`).

22.1 Authentication API (`/auth`)

Method	Endpoint	Auth Required	Description
POST	<code>/auth/login</code>	None	Admin/officer login (OAuth2 form)
POST	<code>/auth/citizen/register</code>	None	Register new citizen
POST	<code>/auth/citizen/login</code>	None	Citizen login, returns JWT
GET	<code>/auth/citizen/profile</code>	CITIZEN JWT	Get citizen profile
PATCH	<code>/auth/citizen/profile</code>	CITIZEN JWT	Update profile fields
GET	<code>/auth/me</code>	Any JWT	Get current user info
POST	<code>/auth/logout</code>	Any JWT	Write audit log entry

Registration Request Body:

```
{
  "name": "Kunal Wandhare",
  "phone": "+919876543210",
  "email": "kunal@example.com",
  "address": "Nagpur, Maharashtra",
  "preferred_language": "en"
}
```

Registration Response:

```
{
  "access_token": "eyJ...",
  "token_type": "bearer",
  "citizen_id": "CIT-001",
  "name": "Kunal Wandhare",
  "message": "Registration successful"
}
```

22.2 Conversation API (/conversation)

Method	Endpoint	Auth Required	Description
POST	/conversation/ message	Optional JWT	Send text message, get AI response
POST	/conversation/ document-upload	Optional JWT	Upload document, get OCR result
POST	/conversation/ voice-message	Optional JWT	Send voice/audio message
POST	/conversation/ resolve-mismatch	Optional JWT	Resolve OCR field mismatch
POST	/conversation/ channel-switch	None	Switch channel (WhatsApp → Web)
GET	/conversation/ session/ {citizen_id}	Optional JWT	Get active session state
POST	/conversation/ admin-doc-decision	Admin JWT	Admin document approve/reject

Message Request:

```
{
  "citizen_identifier": "kunal@example.com",
  "text": "My name is Kunal Wandhare",
  "channel": "WEB",
  "language": "en",
  "modality": "TEXT"
}
```

Message Response:

```
{
  "status": "ok",
  "citizen_ref": "CIT-001",
  "response": "Thank you, Kunal! Now please tell me your date of birth (DD-MM-YYYY).",
  "current_node": "INFORMATION_COLLECTION",
  "filled_slots": { "applicant_name": "Kunal Wandhare" },
  "missing_slots": ["applicant_dob", "annual_income"],
  "application_number": "APP-000001",
  "documents": []
}
```

22.3 Applications API (/applications)

Method	Endpoint	Auth Required	Description
GET	/applications/services	None	List all 4 services from YAML
GET	/applications/services/{service_id}	None	Get service spec (slots, docs, fees)
GET	/applications/my-applications	CITIZEN JWT	Get own applications (ownership enforced)
GET	/applications/citizen/{citizen_ref}	CITIZEN JWT	Get citizen apps with ownership guard
GET	/applications/status/{app_number}	Optional JWT	Get application status + documents

Method	Endpoint	Auth Required	Description
GET	/applications/recent	None	Admin recent applications
PATCH	/applications/status/{app_number}	None	Update status (officer action)
GET	/applications/{app_num}/readiness	Optional JWT	Get 5-component readiness score
GET	/applications/{app_num}/evidence-graph	None	Evidence graph (field → document map)
GET	/applications/{app_num}/certificate	Optional JWT	Download certificate PDF
GET	/applications/current	None	Get non-terminal app (deduplication)
GET	/applications/admin/list	Admin JWT	Admin list with filter/sort/pagination
GET	/applications/admin/{id}	Admin JWT	Admin detail view
POST	/applications/admin/{id}/decision	Admin JWT	APPROVE / REJECT / CLARIFICATION
POST	/applications/validate-eligibility	None	Check eligibility + compute fee
POST	/applications/status/{num}/simulate-approve	None	Dev: simulate approval

Admin Decision Request:

```
{
  "decision": "APPROVE",
  "reason": "All documents verified, income within limit",
  "admin_notes": "Reviewed income certificate and Aadhaar"
}
```

Admin Decision — Processing Steps (from source `applications.py` lines 752–900):

1. Load application by ID/number/tracking_id

2. Validate decision is in {APPROVE, REJECT, REQUEST_CLARIFICATION}
3. if APPROVE: Check current state is in valid approval states, run FSM transition to UNDER_REVIEW → APPROVED → PAYMENT_REQUIRED
4. Persist status updates to SQLite
5. Write AuditLog entry with decision + reason
6. Push notification to citizen's ConversationSession as a chat message
7. broadcast_status_change_sync() to publish SSE event
8. Return new status + citizen notification

22.4 Documents API (/documents)

Method	Endpoint	Auth Required	Description
POST	/documents/upload	Optional JWT	Upload document (multipart/form-data)
GET	/documents/{application_id}	Optional JWT	List application documents
GET	/documents/{doc_id}/fields	Optional JWT	Get OCR extracted + normalized fields
POST	/documents/{doc_id}/resolve-mismatch	Optional JWT	Resolve a field mismatch
DELETE	/documents/{doc_id}	Optional JWT	Delete document

Document Upload Response:

```
{
  "status": "ok",
  "document_id": "uuid-here",
  "doc_type": "IDENTITY_PROOF",
  "ocr_provider": "tesseract",
  "ocr_confidence": 0.90,
  "ocr_fields_extracted": ["applicant_name", "applicant_dob",
"aadhaar_number"],
  "overall_match_score": 87.5,
  "normalized_fields": {
    "applicant_name": "Kunal Wandhare",
    "applicant_dob": "15-06-2000",
    "aadhaar_number": "123456789012"
  }
}
```

```
}
```

22.5 Payment API (/payment)

Method	Endpoint	Auth Required	Description
POST	/payment/initiate	None	Initiate payment (ONLY if PAYMENT_REQUIRED state)
POST	/payment/verify-receipt	None	Upload + OCR-verify payment receipt
GET	/payment/status/{application_id}	None	Get payment status
POST	/payment/simulate-success/{application_id}	None	Dev: instant payment success

Critical Guard (from payment . py lines 50-70):

```
if app.status != AppState.PAYMENT_REQUIRED:
    status_msg = {
        AppState.INITIATED: "Application has not been started
yet.",
        AppState.UNDER_REVIEW: "Application is under review.
Payment enabled after approval.",
        AppState.APPROVED: "Application approved. Payment link will
be sent shortly.",
        ...
    }.get(app.status, f"Payment not available in state:
{app.status}")

    raise HTTPException(
        status_code=400,
        detail=f"Payment is only allowed after government approval.
Current: {app.status}. {status_msg}"
    )
```

Initiate Payment Request:

```
{
  "application_id": "uuid-here",
```

```
"citizen_identifier": "kunal@example.com",
"amount": 50.0,
"channel": "WEB",
"mode": "MOCK_AUTO"
}
```

22.6 Mock Government API (/mock-government)

Method	Endpoint	Auth Required	Description
POST	/mock-government/simulate-decision	Admin JWT	APPROVE / REJECT / CLARIFICATION
POST	/mock-government/submit	None	Submit application (CONSENT_CONFIRMED → UNDER_REVIEW)
GET	/mock-government/status/{tracking_id}	Admin JWT	Get status for review panel

Decision flow (from `mock_government.py` lines 182-195):

- APPROVE: UNDER_REVIEW → APPROVED → PAYMENT_REQUIRED (two transitions, both applied immediately)
- REJECT: UNDER_REVIEW → REJECTED
- CLARIFICATION_REQUIRED: UNDER_REVIEW → CLARIFICATION_REQUIRED

22.7 SSE Streaming API (/stream)

Method	Endpoint	Auth Required	Description
GET	/stream/applications/{id}/events	Optional JWT	Subscribe to application SSE stream
GET	/stream/citizen/{citizen_ref}/events	Optional JWT	Subscribe to citizen SSE stream

Method	Endpoint	Auth Required	Description
GET	/stream/health	None	EventBus health check

SSE Event Format:

```

event: status_change
data: {"type":"status_change","tracking_id":"INC-2026-000001","new_status":"PAYMENT_REQUIRED","actor":"ADMIN","progress":90}

event: heartbeat
data: {"ts":"2026-08-26T10:00:00Z"}

event: connected
data: {"application_id":"uuid","tracking_id":"INC-2026-000001","status":"UNDER_REVIEW","progress":83}

```

22.8 WhatsApp API (/whatsapp)

Method	Endpoint	Auth Required	Description
POST	/whatsapp/webhook	None	Receive incoming WhatsApp message
POST	/whatsapp/message	None	Process WhatsApp message
GET	/whatsapp/session/{identifier}	None	Get session for WhatsApp number
POST	/whatsapp/upload-document	None	Upload document via WhatsApp

22.9 IVR API (/ivr)

Method	Endpoint	Auth Required	Description
POST	/ivr/call/start	None	Start IVR call session
POST	/ivr/call/{call_id}/input	None	Process DTMF/voice input

Method	Endpoint	Auth Required	Description
GET	/ivr/call/{call_id}/status	None	Get IVR call state
POST	/ivr/call/{call_id}/end	None	End IVR call session

22.10 Health & Diagnostics API

Method	Endpoint	Auth Required	Description
GET	/	None	Server info + version
GET	/health	None	Platform health + LLM provider
GET	/health/llm	None	LLM provider name + model
GET	/health/ocr	None	Tesseract version + languages
GET	/api/v1/health/llm	None	Alias

Health Response:

```
{
  "status": "healthy",
  "database": "sqlite",
  "data_guard": "active",
  "llm_provider": "openrouter",
  "timestamp": "2026-08-26T10:00:00Z"
}
```

OCR Health Response:

```
{
  "status": "healthy",
  "available": true,
  "provider": "tesseract",
  "executable": "C:\\Program Files\\Tesseract-OCR\\tesseract.exe",
  "version": "5.3.0",
  "languages": ["eng", "hin", "mar"]
}
```

```
}
```

22.11 Dashboard API (/dashboard)

Method	Endpoint	Auth Required	Description
GET	/dashboard/stats	Admin JWT	Platform statistics
GET	/dashboard/audit	Admin JWT	Recent audit log entries

23. SSE / REAL-TIME EVENT ARCHITECTURE

23.1 EventBus Design

Source: `backend/app/api/routes/stream.py` (341 lines)

The `EventBus` class implements an **in-memory publish-subscribe pattern** using `asyncio.Queue` objects:

```
class EventBus:
    _subscribers: Dict[str, Set[asyncio.Queue]]

    def subscribe(key: str) -> asyncio.Queue
    # Creates a new queue and adds it to _subscribers[key]

    def unsubscribe(key: str, queue: asyncio.Queue)
    # Removes queue from _subscribers[key]

    async def publish(key: str, event: dict)
    # Puts event into all queues for key (async)

    def publish_sync(key: str, event: dict)
    # Puts event into all queues for key (sync, from non-async
code)
```

```
def subscriber_count(key: str) -> int
# Returns number of active connections for a key
```

23.2 Dual-Key Publishing

When an admin makes a decision, SSE events are published to TWO keys:

```
# From mock_government.py _broadcast_sse()
broadcast_status_change_sync(
    application_id=tracking_id,    # Application-level subscribers
    ...
)
bus.publish_sync(citizen_ref, event) # Citizen-level subscribers
```

This ensures that:

- The `CitizenChat` page (subscribed to `citizen_ref`) gets the notification
- The `AdminDashboard` (subscribed to `application_id`) also gets the update

23.3 Heartbeat Mechanism

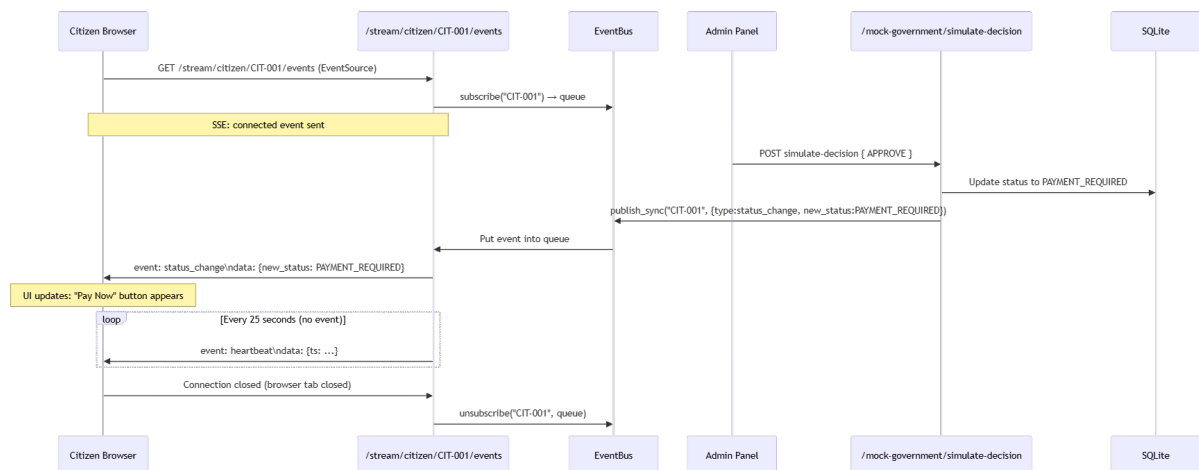
The SSE endpoint sends a heartbeat every 25 seconds to keep the connection alive through proxies and load balancers. If no event is published within 20 seconds, a heartbeat is sent:

```
# In stream.py SSE generator
try:
    event = await asyncio.wait_for(queue.get(), timeout=20.0)
    yield f"event: {event['type']}\ndata: {json.dumps(event)}\n\n"
except asyncio.TimeoutError:
    yield f"event: heartbeat\ndata: {json.dumps({'ts':
datetime.utcnow().isoformat()})}\n\n"
```

23.4 Citizen Isolation

A citizen's SSE channel is keyed by their `citizen_ref` (e.g., "CIT-001"). The backend validates that the JWT token's `citizen_ref` matches the requested subscription key before allowing the connection. Admin and Officer roles can subscribe to any application or citizen.

23.5 SSE Event Flow



24. CITIZEN WORKFLOW

24.1 Complete Citizen Journey

```

STEP 1: REGISTRATION
Citizen visits /login -> AuthPage
Enters name, phone, email, password
POST /auth/citizen/register
├─ Creates Citizen record (citizen_ref: CIT-001)
├─ Creates 3 ChannelIdentity records (SHA-256 hashed)
└─ Returns JWT -> stored in authStore (localStorage)

STEP 2: START APPLICATION
Citizen navigates to /assistant -> CitizenChat
Types: "I want an income certificate" or clicks service tile
POST /conversation/message
├─ ConversationOrchestrator detects SERVICE_SELECTED intent
├─ Creates Application record in SQLite (status: SERVICE_SELECTED)
└─ FSM -> INFORMATION_COLLECTION
AI Response: "Welcome! For Income Certificate, I need some
  
```

details.

Please tell me your full name."

STEP 3: CONSENT

AI asks for terms acceptance

Citizen: "Yes, I agree"

- session.consent_given = True
- FSM → CONSENT_GIVEN

STEP 4: SLOT FILLING (INFORMATION_COLLECTION)

AI guides through required fields (per YAML service spec):

- Full Name → validated against anti-fake-name guard
- Date of Birth → validated as DD-MM-YYYY
- Annual Income → validated as number within bounds
- Caste Category → validated against allowed_values
- Address, Phone, etc.

Each value stored as ApplicationData (encrypted if RESTRICTED)

Progress: 25% → 35% as fields are collected

STEP 5: DOCUMENT UPLOAD

FSM → DOCUMENT_COLLECTION

AI: "Please upload your Aadhaar card."

Citizen uploads file via drag-drop / file picker

POST /conversation/document-upload (multipart)

- File saved to data/uploads/
- OCRService.run_ocr() → extract 10+ fields
- MatchingService compares to declared fields
 - MATCHED: document verified ✓
 - MISMATCH: citizen prompted to resolve
- Document stored in documents table

STEP 6: OCR MISMATCH RESOLUTION (if needed)

If OCR extracted name "Kunal W." but declared "Kunal Wandhare":

AI shows: "Your Aadhaar shows 'Kunal W.' but you entered 'Kunal Wandhare'.

Which is correct? [Use Document] [Keep My Answer]
[Enter Manually]"

Citizen picks resolution → POST /conversation/resolve-mismatch

STEP 7: READINESS CHECK

FSM → READINESS_CHECK

GET /applications/{num}/readiness

- Field Completeness: 30 pts (all required fields present)
- Document Coverage: 25 pts (all required docs uploaded)
- OCR Validation: 20 pts (OCR matched declared values)
- Eligibility: 15 pts (income within limit, age ≥ 18)
- Cross-field Consistency: 10 pts (DOB valid, income > 0)

Score ≥ 75 AND no blocking issues → READY_FOR_REVIEW

Score < 75 → FIX_REQUIRED (AI guides citizen to fix)

STEP 8: FINAL REVIEW & SUBMISSION

Citizen reviews all collected data

Citizen clicks "Submit Application"

POST /mock-government/submit

- FSM: CONSENT_CONFIRMED → SUBMITTED → UNDER_REVIEW
- SSE event: {type: status_change, new_status: UNDER_REVIEW}
- UI: "Your application is submitted and under government review."

STEP 9: WAITING (UNDER_REVIEW)

Browser SSE connection is open

```
Admin reviews application in admin panel
When admin approves → SSE pushes event instantly
```

```
STEP 10: PAYMENT (after APPROVAL)
SSE event: {new_status: PAYMENT_REQUIRED}
UI: "PAY NOW" button appears (₹50)
Citizen clicks Pay Now
POST /payment/initiate { mode: MOCK_AUTO }
├─ Guard checks app.status == PAYMENT_REQUIRED (HTTP 400 if not)
├─ PaymentService.initiate_payment()
└─ MOCK_AUTO: immediately marks as SUCCESS
FSM → PAYMENT_COMPLETED → CERTIFICATE_GENERATION
```

```
STEP 11: CERTIFICATE
CertificateService generates PDF
FSM → CERTIFICATE_READY → COMPLETED
GET /applications/{num}/certificate → FileResponse PDF
Citizen downloads certificate 📄
```

24.2 Citizen Workflow Diagram

```
graph TD
    REG["Registration<br>/auth/citizen/register"] --> LOGIN["Login<br>/auth/citizen/login"]
    LOGIN --> CHAT["CitizenChat<br>/assistant"]
    CHAT --> CONSENT["Consent<br>session.consent_given=True"]
    CONSENT --> SVC["Service Selection<br>income/caste/domicile/obc"]
    SVC --> FILL["Slot Filling<br>(AI-guided, field-by-field)"]
    FILL --> |"All fields done"| UPLOAD["Document Upload<br>Aadhaar, Income Proof, etc."]
    UPLOAD --> OCR["OCR Processing<br>Tesseract + PyMuPDF"]
    OCR --> MATCH["Match Score?"]
    MATCH --> |"Matched"| READY["Readiness Check<br>Score 0-100"]
    MATCH --> |"Mismatch"| RESOLVE["Citizen Resolves Mismatch<br>/resolve-mismatch"]
    RESOLVE --> READY
    READY --> |"> 75"| REVIEW["Final Review<br>Citizen confirms all data"]
    READY --> |"< 75"| FIX["Fix Required<br>AI guides correction"]
    FIX --> FILL
    REVIEW --> SUBMIT["Submit<br>/mock-government/submit"]
    SUBMIT --> WAIT["Waiting: UNDER_REVIEW<br>(SSE connection open)"]
    WAIT --> |"SSE: APPROVED"| PAY["Payment ₹50<br>/payment/initiate"]
    WAIT --> |"SSE: REJECTED"| DONE_REJ["Application Rejected ❌"]
    WAIT --> |"SSE: CLARIFICATION"| FILL
    PAY --> CERT["Certificate Generated<br>PDF Download"]
    CERT --> DONE["✅ COMPLETED"]
```

25. ADMIN / OFFICER WORKFLOW

25.1 Admin Review Flow

Source: `api/routes/applications.py`, `api/routes/mock_government.py`

```
STEP 1: ADMIN LOGIN
  Admin navigates to /admin/login → AdminLogin page
  POST /auth/login (form data: username, password)
  |— Validates against User table (role: ADMIN)
  |— Returns JWT with role: ADMIN
  |— authStore.setAuth(token, user)

STEP 2: DASHBOARD
  /admin/dashboard → AdminDashboard
  GET /dashboard/stats
  Shows: total applications, pending review, approved today,
  rejected today

STEP 3: APPLICATION QUEUE
  /admin/applications → AdminApplications
  GET /applications/admin/list?
status=UNDER_REVIEW&sort_by=newest&page=1
  Shows filterable, sortable list with:
  |— Application number, tracking ID
  |— Service type, citizen ref
  |— Current status, progress %
  |— Readiness score (from ReadinessEngine)
  |— Overall document match score
  |— Created/submitted timestamps

STEP 4: OPEN APPLICATION FOR REVIEW
  Admin clicks application → /admin/review/{id}
  GET /applications/admin/{id}
  Page shows 7 sections:
  |— Section 1: Application Overview (tracking ID, service,
status, timestamps)
  |— Section 2: Personal Data (all ApplicationData fields,
decrypted)
  |— Section 3: Documents & OCR (uploaded docs + raw + normalized
OCR)
  |— Section 4: Readiness Score (5-component breakdown with bars)
  |— Section 5: Document Matching (field-by-field match scores)
  |— Section 6: Fraud & Risk Review (anomaly score, Data Guard
log)
  |— Section 7: Decision Panel (Approve / Reject / Request
Clarification)

STEP 5: MAKE DECISION
```

```

    Admin clicks "Approve" and confirms
    POST /applications/admin/{id}/decision { decision: "APPROVE",
reason: "..."}
    |— Validate FSM: current state must be in valid_approval_states
    |— FSM transitions: → UNDER_REVIEW → APPROVED → PAYMENT_REQUIRED
    |— AuditLog written: event_type=GOVERNMENT_DECISION,
outcome=SUCCESS
    |— Chat notification pushed to citizen's ConversationSession
    |— SSE broadcast: publish_sync(tracking_id) +
publish_sync(citizen_ref)

    STEP 6: CITIZEN IS NOTIFIED INSTANTLY
    Citizen's browser receives SSE event
    UI updates: "PAY NOW" button appears
    CitizenChat shows: "🎉 Your application has been APPROVED! Please
pay ₹50."

```

25.2 Admin Decision Logic

From `applications.py` `submit_admin_decision_endpoint()`:

```

if decision == "APPROVE":
    # Only allowed from specific states
    valid_approval_states = (
        AppState.SUBMITTED_FOR_VERIFICATION,
        AppState.UNDER_REVIEW,
        AppState.READY_FOR_REVIEW,
        "FINAL_REVIEW",
        "CONSENT_CONFIRMED",
    )

    # Run FSM transitions
    fsm.transition(AppState.UNDER_REVIEW)
    repo.update_status(app.id, AppState.UNDER_REVIEW)
    # ... continues to APPROVED → PAYMENT_REQUIRED

elif decision == "REJECT":
    fsm.transition(AppState.REJECTED)
    repo.update_status(app.id, AppState.REJECTED)
    # TERMINAL STATE

elif decision in ("REQUEST_CLARIFICATION",
"CLARIFICATION_REQUIRED"):
    fsm.transition(AppState.CLARIFICATION_REQUIRED)
    repo.update_status(app.id, AppState.CLARIFICATION_REQUIRED)
    # Citizen can respond and resubmit

```

26. DOCUMENT VERIFICATION WORKFLOW

26.1 Required Documents per Service

Documents are defined in each YAML service spec (`seed/service_specs/*.yaml`):

Service	Required Documents
Income Certificate	Identity Proof (Aadhaar/Voter ID), Income Proof (Salary Slip/Bank Statement)
Caste Certificate	Identity Proof, Existing Caste Proof (if available)
OBC-NCL Certificate	Identity Proof, Income Proof, Caste/Category Proof
Domicile Certificate	Identity Proof, Residence Proof (utility bill/rent agreement)

26.2 Document Upload Flow

```
Citizen Uploads File
|
v
/conversation/document-upload (multipart/form-data)
├── citizen_identifier, doc_type, channel, file
└──
|
v
File saved: data/uploads/{citizen_ref}_{filename}
|
v
OCRService.run_ocr(file_path, doc_type)
├── Extract text (Tesseract/PyMuPDF)
├── Detect document type
├── Extract 10 field types
├── Normalize all fields
└── Optional: LLM normalization (if DataGuard allows)
|
v
MatchingService.match(ocr_fields, declared_fields)
├── Compare each OCR field to its declared counterpart
├── Score each field (0-100, exact or fuzzy)
├── Identify mismatches
└── Compute overall_match_score
|
v
```

```

DocumentRepository.create()
├── Store: file_ref, doc_type, raw_ocr_text, raw_extracted_fields,
    normalized_fields, field_match_scores,
overall_match_score,
    mismatch_fields, verification_status
├── verification_status: PENDING → MATCHED (score ≥ 80) or MISMATCH
└──
    ▼
Return to frontend:
├── OCR provider + confidence
├── Extracted field list
├── Normalized fields
├── Overall match score
└── Updated document list with verification status

```

26.3 Mismatch Resolution

When OCR detects a mismatch (e.g., OCR extracted "Kunal W." but declared "Kunal Wandhare"), the citizen is offered three resolution options:

1. **use_document**: Accept the OCR value, update `filled_slots` with document value
2. **use_declared**: Keep the declared value, mark mismatch as resolved
3. **manual_entry + new_value**: Citizen types a corrected value

Source: `/conversation/resolve-mismatch → _handle_mismatch_resolution()` in orchestrator.

27. OCR NORMALIZATION & MATCHING WORKFLOW

27.1 Normalization Pipeline Detail

The goal of normalization is to convert noisy OCR output into clean, comparable values:

Field	Raw OCR Example	After Normalization
applicant_name	"Ee Ae D Viki Bhausahab Lokhande Oa Bs Ga"	"Viki Bhausahab Lokhande"
applicant_dob	"15/06/2000"	"15-06-2000"
applicant_dob	"2000-06-15"	"15-06-2000"
annual_income	"₹ 3,00,000"	"300000"
aadhaar_number	"1234 5678 9012"	"123456789012"
pan_number	"abcde1234f"	"ABCDE1234F"
gender	"पुरुष"	"Male"

27.2 Matching Algorithm

Source: `services/matching_service.py` (20 KB):

The matching service compares normalised OCR fields against declared application fields:

```

For each required field:
  — Get declared value from ApplicationData
  — Get normalised OCR value from Document.normalized_fields
  — Exact match check (case-insensitive after further
normalisation)
    → 100% score
  — Fuzzy match check (for names: Levenshtein or token sort ratio)
    → Score 0-99%
  — Date normalisation match (DD-MM-YYYY canonical form)
  — Income match: strip ₹, commas, parse as int
  — Overall score = weighted average of all field scores
    verified: score ≥ 80 → MATCHED
    unverified: score < 80 → MISMATCH (citizen must resolve)

```

27.3 Evidence Graph

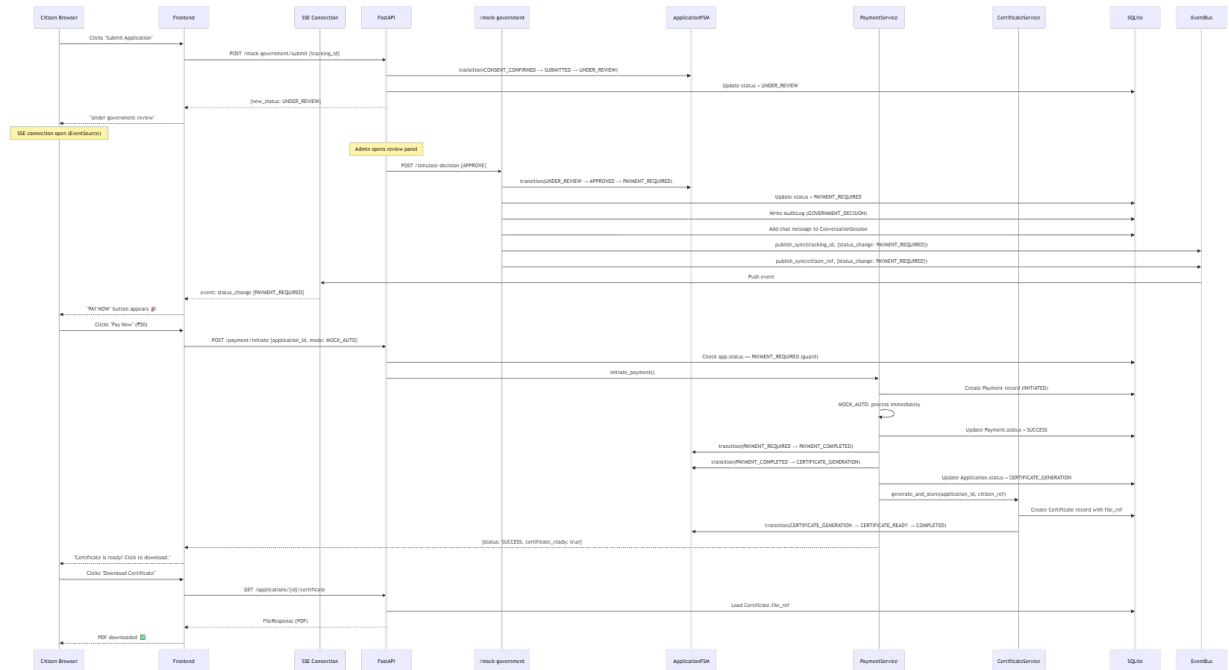
Source: `applications.py /evidence-graph` endpoint:

For admin review, the evidence graph shows which document verified which field:

```
{
  "fields": {
    "applicant_name": {
      "declared_value": "Kunal Wandhare",
      "doc_sources": ["IDENTITY_PROOF"],
      "verified": true,
      "conflicting": false,
      "confidence": 100
    },
    "annual_income": {
      "declared_value": "300000",
      "doc_sources": ["INCOME_PROOF"],
      "verified": true,
      "conflicting": false,
      "confidence": 95
    }
  },
  "summary": {
    "total_fields": 6,
    "verified_fields": 5,
    "conflicting_fields": 0,
    "verification_coverage_pct": 83.3
  }
}
```

28. APPROVAL → PAYMENT → CERTIFICATE WORKFLOW

28.1 Full Post-Submission Sequence



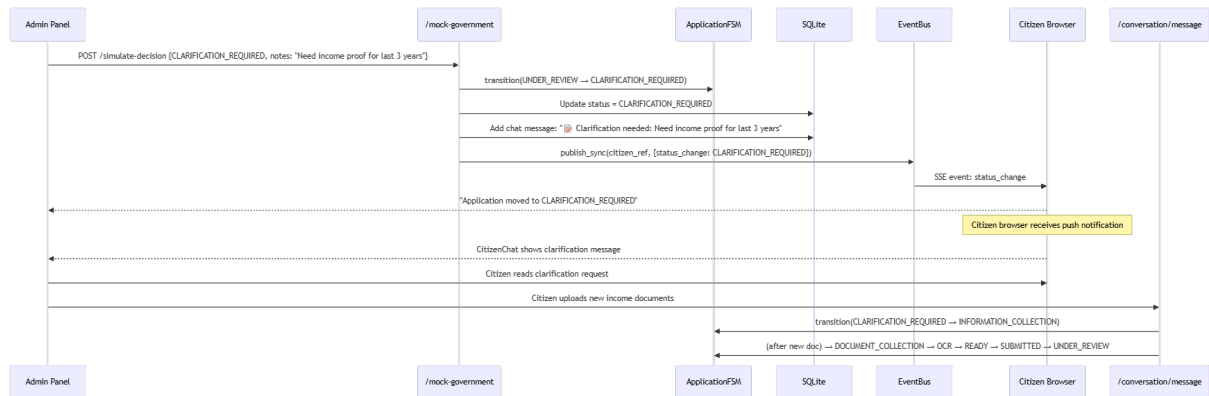
28.2 Payment State Guard Detail

The payment endpoint (`/payment/initiate`) enforces strict state validation. Calling it in any state other than `PAYMENT_REQUIRED` returns HTTP 400 with a specific message:

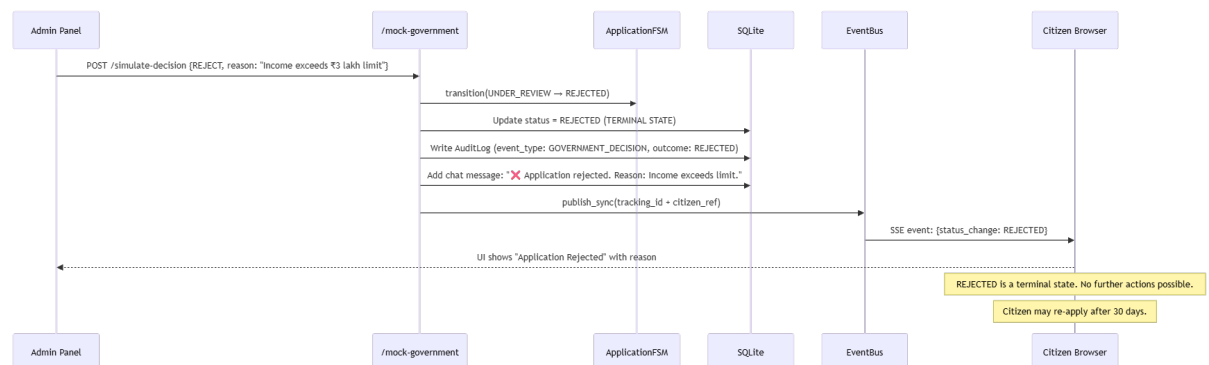
Current State	Error Message
INITIATED	"Application has not been started yet."
INFORMATION_COLLECTION	"Application is still collecting information."
UNDER_REVIEW	"Under government review. Payment enabled after approval."
APPROVED	"Application approved. Payment link sent shortly."
PAYMENT_COMPLETED	"Payment already completed."
REJECTED	"Application rejected. Payment not applicable."

29. CLARIFICATION & REJECTION WORKFLOWS

29.1 Clarification Flow



29.2 Rejection Flow



30. READINESS ENGINE

30.1 Five-Component Scoring

Source: `backend/app/services/readiness_engine.py` (525 lines)

```

    ReadinessEngine.compute(
        service_id,      ← which service (income/caste/etc.)
        filled_slots,    ← dict of field: value
        required_slots,  ← list of required field names from YAML
        required_docs,   ← list of required doc types from YAML
        uploaded_docs,   ← list of doc_type strings from database
        ocr_results,     ← list of {doc_type, status,
overall_match_score}
        eligibility_result ← {eligible: bool, reason: str}
    )

```

Component 1: Field Completeness (max 30 pts)

```

score = (count of filled required fields / total required fields) ×
30

```

Component 2: Document Coverage (max 25 pts)

```

score = (count of required docs that are uploaded / total required
docs) × 25

```

Component 3: OCR Validation (max 20 pts)

```

ocr_coverage = uploaded_docs_with_ocr / required_docs_count
avg_match = mean(all doc.overall_match_score) / 100
score = (0.5 × ocr_coverage + 0.5 × avg_match) × 20

```

Component 4: Eligibility (max 15 pts)

```

if eligibility_result.eligible == True:  score = 15
if eligibility_result.eligible == False: score = 0
if eligibility_result is None:           score = 10.5 (70% of 15)

```

Component 5: Cross-field Consistency (max 10 pts)

Runs 5 consistency checks:

- 1. Mobile number format: `^[6-9]\d{9}$`
- 2. DOB is a past date (not future)
- 3. Annual income > 0 (parseable as float)
- 4. Family income ≥ individual income (if both present)
- 5. Aadhaar = exactly 12 digits

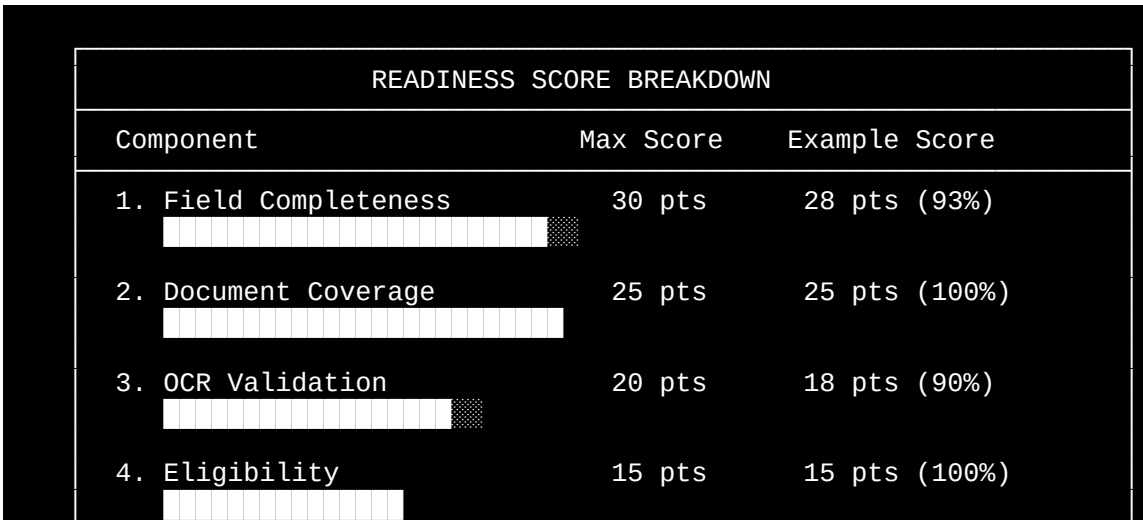
$$\text{score} = (\text{checks_passed} / \text{total_checks}) \times 10$$

Status Thresholds:

Total Score	Status	Can Submit
≥ 90	READY	✔ Yes
≥ 75	MINOR_ISSUES	✔ Yes
≥ 60	MODERATE_ISSUES	✘ No
< 60	MAJOR_ISSUES	✘ No

`can_submit = (overall_score >= 75.0) AND (len(blocking_issues) == 0)`

30.2 Readiness Score Diagram



5. Cross-field Consistency	10 pts	8 pts (80%)
TOTAL SCORE	100 pts	94 pts
STATUS: READY 		
CAN SUBMIT: YES		

31. PAYMENT ARCHITECTURE

31.1 Payment Flow

Source: `backend/app/services/payment_service.py` (~12 KB) and `backend/app/api/routes/payment.py` (149 lines)

```

POST /payment/initiate
├── Validate app.status == PAYMENT_REQUIRED (HTTP 400 if not)
├── Check existing payment not PAID/SUCCESS
├── CitizenRepository.resolve_or_create(citizen_identifier)
├── PaymentService.initiate_payment(
│   application_id, amount, citizen_ref, channel, mode
│ )
├── Create Payment record: status=INITIATED
├── If mode == "MOCK_AUTO":
│   └── _process_mock() → immediately set SUCCESS
├── If mode == "UPI_QR":
│   └── Return UPI QR code data for citizen to scan
├── On SUCCESS:
│   ├── Update Application.payment_status = SUCCESS
│   ├── FSM: PAYMENT_REQUIRED → PAYMENT_COMPLETED
│   ├── FSM: PAYMENT_COMPLETED → CERTIFICATE_GENERATION
│   ├── Trigger CertificateService.generate_and_store()
│   └── FSM: CERTIFICATE_GENERATION → CERTIFICATE_READY →
COMPLETED
└── Return {status, transaction_id, amount, certificate_url}

```

31.2 Payment Modes

Mode	Description
MOCK_AUTO	Development mode. Payment is instantly marked SUCCESS. No real money.
UPI_QR	Production mode. Returns UPI QR code. Citizen scans and pays. Receipt required.

31.3 Receipt Verification

```

POST /payment/verify-receipt (multipart)
├── Save uploaded receipt file (image/PDF)
├── OCRService.run_ocr(receipt_path, "PAYMENT_RECEIPT")
│   └── Extracts: transaction_id, amount, payment status, date
├── PaymentService.verify_receipt(application_id, file_path)
│   ├── Cross-reference transaction_id against Payment records
│   ├── Verify amount matches expected fee
│   └── If verified → mark Payment.status = SUCCESS
└── Trigger certificate generation

```

32. CERTIFICATE GENERATION

32.1 Certificate Service Flow

Source: `backend/app/services/certificate_service.py`

```

CertificateService.generate_and_store(application_id, citizen_ref)
├── Load Application + ApplicationData (decrypt all RESTRICTED fields)
├── Generate certificate content:
│   ├── Header: Government of Maharashtra Revenue Department
│   ├── Certificate number: CERT-{YYYYY}-{padded_id}
│   ├── Citizen name, service type
│   ├── Official declaration text
│   ├── Date of issue
│   └── Digital signature placeholder

```

```

— Save PDF to: data/certificates/{certificate_number}.pdf
— Create Certificate record in SQLite:
  — certificate_number (UNIQUE)
  — file_ref (absolute path)
  — issue_date
  — digital_signature
— Update Application.status = CERTIFICATE_READY

```

32.2 Certificate Download

```

GET /applications/{id_or_number}/certificate
— Load Application by ID, number, or tracking_id
— Verify JWT ownership (citizen must own this application)
— If certificate.file_ref exists and file present:
  — Return FileResponse(path, media_type="application/pdf",
                        filename="Certificate_INC-2026-
0000001.pdf")
— If certificate not yet generated:
  — CertificateService.generate_and_store() on demand
  — Return PDF
— If still missing: HTTP 404 "Certificate has not been issued
yet"

```

33. SECURITY ARCHITECTURE

33.1 Security Layers

```

Layer 1: Transport Security
- HTTPS enforcement (production) [Not verified in source code]
- CORS whitelist: localhost ports 3000, 5173, 5174, 8000

Layer 2: Application Firewall (SecurityMiddleware)
- Rate limiting on all endpoints
- Security headers: X-Content-Type-Options, X-Frame-Options,
  X-XSS-Protection, Referrer-Policy, Permissions-Policy

Layer 3: Authentication (JWT)
- Algorithm: HS256

```

- Expiry: 8 hours
- Secret: SECRET_KEY (env var, must be changed in production)

Layer 4: Authorization (FastAPI Depends)

- get_current_citizen() → enforces CITIZEN role
- require_admin() → enforces ADMIN/OFFICER role
- verify_application_ownership() → citizen_ref ownership check

Layer 5: Data Protection

- AES-256-GCM encryption for all RESTRICTED/QUASI_IDENTIFIER fields
- SHA-256 hashing for all channel identifiers
- No raw phone/email in channel_identities table

Layer 6: AI Trust Boundary

- DataGuard.check() on ALL outbound LLM payloads
- RESTRICTED_FIELDS set (18 fields) → immediate BLOCK
- k-anonymity check for quasi-identifiers
- Every BLOCK decision logged to AuditLog

Layer 7: Payment Guard

- app.status == PAYMENT_REQUIRED check before any payment
- Prevents payment before government approval
- Prevents double payment

Layer 8: FSM Guard

- ApplicationFSM.can_transition_to() validates all state changes
- No direct database status updates without FSM validation

Layer 9: Audit Trail

- Append-only AuditLog table
- SHA-256 hash chaining (previous_hash)
- Every decision, consent, payment, and Data Guard action logged

33.2 AES-256-GCM Field Encryption Detail

Source: `backend/app/data_layer/encryption.py`

```
class FieldEncryptor:
    @classmethod
    def encrypt(cls, plaintext: str) -> str:
        key = _derive_key() # SHA-256(SECRET_KEY) → 32
        bytes
        aesgcm = AESGCM(key)
        nonce = os.urandom(12) # 96-bit random nonce per
        encryption
        ciphertext = aesgcm.encrypt(nonce, plaintext.encode("utf-
8"), None)
        # Store format: base64(nonce[12 bytes] || ciphertext)
        return base64.b64encode(nonce + ciphertext).decode("ascii")
```

```

@classmethod
def decrypt(cls, encrypted: str) -> str:
    key = _derive_key()
    aesgcm = AESGCM(key)
    raw = base64.b64decode(encrypted.encode("ascii"))
    nonce, ciphertext = raw[:12], raw[12:]
    return aesgcm.decrypt(nonce, ciphertext, None).decode("utf-8")

@classmethod
def hash_token(cls, value: str) -> str:
    # HMAC-SHA256 for generating citizen_ref from phone
    key = _derive_key()
    return hmac.new(key, value.encode(), sha256).hexdigest()
[:32]

```

Every applicant_name, aadhaar_number, pan_number, applicant_dob, phone_number, email_address, and address field in ApplicationData is stored as an AES-256-GCM encrypted string. The plaintext never touches SQLite.

34. PRIVACY & PII PROTECTION

34.1 PII Classification Table

Field	Classification	Storage	LLM Allowed	OCR Extracted
aadhaar_number	RESTRICTED	AES-256-GCM encrypted	✗ NEVER	✓ (for matching)
pan_number	RESTRICTED	AES-256-GCM encrypted	✗ NEVER	✓ (for matching)
applicant_name	RESTRICTED	AES-256-GCM encrypted	✗ NEVER	✓ (for matching)
applicant_dob	RESTRICTED	AES-256-GCM encrypted	✗ NEVER	✓ (for matching)
phone_number	RESTRICTED	AES-256-GCM encrypted	✗ NEVER	—
email_address	RESTRICTED	AES-256-GCM encrypted	✗ NEVER	—

Field	Classification	Storage	LLM Allowed	OCR Extracted
address	RESTRICTED	AES-256-GCM encrypted	✗ NEVER	✓ (for matching)
father_name	RESTRICTED	AES-256-GCM encrypted	✗ NEVER	✓ (for matching)
mother_name	RESTRICTED	AES-256-GCM encrypted	✗ NEVER	—
passport_number	RESTRICTED	AES-256-GCM encrypted	✗ NEVER	—
annual_income	QUASI_IDENTIFIER	AES-256-GCM encrypted	Synthetic only	✓
caste_category	QUASI_IDENTIFIER	AES-256-GCM encrypted	Synthetic only	✓
district	QUASI_IDENTIFIER	AES-256-GCM encrypted	Synthetic only	—
occupation	QUASI_IDENTIFIER	AES-256-GCM encrypted	Synthetic only	—
service_id	NON_SENSITIVE	Plain text	✓	—
application_id	NON_SENSITIVE	Plain text	✓	—
tracking_id	NON_SENSITIVE	Plain text	✓	—
status	NON_SENSITIVE	Plain text	✓	—

34.2 Channel Identity Privacy

When a citizen registers with phone +919876543210:

- The phone is stored as SHA-256("+919876543210") in `channel_identities.identifier_hash`
- The raw phone is stored encrypted in `citizens.phone_encrypted` (AES-256-GCM)
- The `citizen_ref` (e.g., "CIT-001") is used for all cross-table linkage
- No table in the system stores an unencrypted phone number alongside a citizen name

35. LOGGING & AUDIT ARCHITECTURE

35.1 Application Logging

The platform uses Python's standard `logging` module with `struct log` for structured output:

```
# Configured in main.py
logging.basicConfig(
    level=logging.DEBUG if settings.DEBUG else logging.INFO,
    format="%(asctime)s | %(levelname)-8s | %(name)s | %(message)s",
    stream=sys.stdout,
)
```

Each service module uses its own named logger:

```
logger = logging.getLogger(__name__) # e.g.,
"app.services.ocr_service"
```

Key log events (verified from source):

- OCR: `[OCR_START]`, `[OCR_COMPLETED]`, `[OCR_EMPTY]`, `[OCR_AI_NORM_SKIPPED]`
- Data Guard: `[DATA_GUARD_BLOCK]`, `[DATA_GUARD_ALLOW]`
- FSM: State transitions logged at INFO level
- Payment: INITIATED, SUCCESS, FAILED logged with transaction IDs
- Admin decisions: All APPROVE/REJECT/CLARIFICATION at INFO level

35.2 Immutable Audit Log

The `AuditLog` table provides a tamper-evident, append-only audit trail:

```

class AuditRepository:
    def write(
        self,
        event_type: str,          # DATA_GUARD | CONSENT | PAYMENT |
SUBMISSION | AUTH | GOVERNMENT_DECISION
        actor: str,              # Username, "CITIZEN",
"SYSTEM_SIMULATOR", "ADMIN"
        citizen_ref: str,
        application_id: str,
        action: str,             # Human-readable description
        outcome: str,            # ALLOW | BLOCK | SUCCESS | FAILURE
        blocked_fields: List[str] = [],
        metadata: Dict = {},
        payload_hash: str = None,
    )
    # Each write:
    # 1. Computes SHA-256 of key fields
    # 2. Fetches previous_hash from last AuditLog entry
    # 3. Inserts new row (auto-increment integer PK)
    # 4. NEVER updates existing rows

```

Event Types Logged:

Event Type	Logged When
DATA_GUARD	Every DataGuard check (ALLOW or BLOCK)
CONSENT	Citizen gives/revokes consent
PAYMENT	Payment initiated, completed, or failed
SUBMISSION	Application submitted for verification
AUTH	Login, registration, logout
GOVERNMENT_DECISION	Admin approves/rejects/clarifies
STATUS_UPDATE	Officer updates application status
ADMIN_DOC_DECISION	Admin document pre-payment decision

35.3 Audit Log Architecture

```

graph TD
    APP["Application Events<br/>(Login, Status, Payment)"]

```

```

DG["DataGuard Decisions<br/>(ALLOW/BLOCK)"]
ADMIN["Admin Decisions<br/>(APPROVE/REJECT)"]

AUDIT["AuditRepository.write()"]

subgraph LOG["AuditLog Table (Append-Only)"]
  E1["Entry 1<br/>hash: 'abc123'<br/>prev_hash: NULL"]
  E2["Entry 2<br/>hash: 'def456'<br/>prev_hash: 'abc123'"]
  E3["Entry 3<br/>hash: 'ghi789'<br/>prev_hash: 'def456'"]
  E4["Entry N<br/>hash: 'xyz999'<br/>prev_hash: 'ghi789'"]
end

APP & DG & ADMIN --> AUDIT
AUDIT --> E1 --> E2 --> E3 --> E4

VERIFY["Chain Verifier<br/>SHA-256 check"]
E4 --> VERIFY

style AUDIT fill:#fcc419,color:#000

```

36. DEPLOYMENT ARCHITECTURE

36.1 Local Development Deployment

Verified from source code and README:

```

Developer Machine
├── Terminal 1: Backend
│   ├── cd backend
│   ├── pip install -r requirements.txt
│   ├── uvicorn main:app --host 0.0.0.0 --port 8000 --reload
│   ├── → http://localhost:8000
│   └── → http://localhost:8000/docs (Swagger UI)
├── Terminal 2: Frontend
│   ├── cd frontend
│   ├── npm install
│   ├── npm run dev
│   └── → http://localhost:5173 (Vite dev server)
├── Local Services:
│   ├── SQLite: backend/revenue_services.db (auto-created)
│   ├── Tesseract: C:\Program Files\Tesseract-OCR\tesseract.exe
│   └── OpenRouter/Gemini/Groq: accessed via HTTPS from developer
(Windows)

```

machine

36.2 Production Architecture

Note: Production deployment configuration is NOT verified in source code. The following is based on the project's design intent and technology choices.

```
(Not verified in source code)

Internet
  |
  ▼
Load Balancer / Reverse Proxy (Nginx or Cloudflare)
  |
  ├── Frontend: Vite production build → static file server
  |   |   npm run build → dist/ directory
  |
  └── Backend: uvicorn behind gunicorn
      |   uvicorn main:app --host 0.0.0.0 --port 8000 --workers 4
      |   └── SQLite (single-file) OR PostgreSQL (production
recommended)
```

36.3 Environment Separation

```
Development:
  DEBUG=True
  DATABASE_URL=sqlite:///backend/revenue_services.db
  LLM_PROVIDER=openrouter (free tier)
  PAYMENT_ADAPTER=mock
  NOTIFICATION_ADAPTER=mock

Production (target – not verified):
  DEBUG=False
  SECRET_KEY=<strong-random-key-never-hardcoded>
  ADMIN_PASSWORD=<strong-password>
  DATABASE_URL=<production-database-url>
  LLM_PROVIDER=openrouter
  OPENROUTER_API_KEY=<production-key>
  PAYMENT_ADAPTER=<real-gateway>
  DATA_GUARD_ENABLED=True ← MUST be True
  DATA_GUARD_LOG_ALL=True ← MUST be True for compliance
```

37. CONFIGURATION MANAGEMENT

37.1 Environment Variables Reference

Source: `backend/app/core/config.py`

Core Application:

Variable	Type	Default	Required
APP_NAME	str	"Multilingual AI-Powered Citizen Revenue Services Platform"	No
APP_VERSION	str	"3.0.0"	No
DEBUG	bool	False	No
SECRET_KEY	str	(dev key)	Yes — change in production
API_V1_PREFIX	str	"/api/v1"	No

Database:

Variable	Type	Default	Required
DATABASE_URL	str	"sqlite:///backend/revenue_services.db"	No

LLM Provider (exactly ONE set required):

Variable	Type	Default	Notes
LLM_PROVIDER	str	"openrouter"	"openrouter" "gemini" "groq"
OPENROUTER_API_KEY	str	None	Required if LLM_PROVIDER=openrouter
OPENROUTER_MODEL	str	"minimax/minimax-m3:free"	Any OpenRouter model
GEMINI_API_KEY	str	None	Required if LLM_PROVIDER=gemini
GEMINI_MODEL	str	"gemini-1.5-flash"	—
GROQ_API_KEY	str	None	Required if LLM_PROVIDER=groq
GROQ_MODEL	str	"llama3-8b-8192"	—
LLM_MAX_TOKENS	int	1000	Max completion tokens

Data Guard:

Variable	Type	Default	Notes
DATA_GUARD_ENABLED	bool	True	NEVER set False in production
DATA_GUARD_LOG_ALL	bool	True	Log ALLOW decisions too

OCR:

Variable	Type	Default	Notes
TESSERACT_CMD	str	Windows path	Path to tesseract binary
TESSERACT_PATH	str	Windows path	Alias for TESSERACT_CMD

Security & Auth:

Variable	Type	Default	Notes
JWT_ACCESS_TOKEN_EXPIRE_MINUTES	int	480	8-hour expiry
ADMIN_USERNAME	str	"admin"	Seeded admin username
ADMIN_PASSWORD	str	"Admin@123"	Change in production

Storage Paths:

Variable	Default
STORAGE_PATH	backend/data/uploads
RECEIPT_PATH	backend/data/receipts
CERTIFICATE_PATH	backend/data/certificates
AUDIO_PATH	backend/data/audio

Adapters:

Variable	Default	Options
PAYMENT_ADAPTER	"mock"	"mock" real gateway
NOTIFICATION_ADAPTER	"mock"	"mock" real SMS/WhatsApp
DOCUMENT_ADAPTER	"mock"	"mock" cloud storage
AUTH_ADAPTER	"mock"	"mock" production

Fraud Detection:

Variable	Default	Notes
FRAUD_SCORE_THRESHOLD_REVIEW	0.4	Flag for manual review
FRAUD_SCORE_THRESHOLD_REJECT	0.7	Auto-reject threshold

37.2 Sample .env File

```
# — LLM Provider —————
LLM_PROVIDER=openrouter
OPENROUTER_API_KEY=sk-or-v1-your-real-api-key-here
OPENROUTER_MODEL=minimax/minimax-m3:free

# — Database —————
DATABASE_URL=sqlite:///backend/revenue_services.db

# — Security —————
SECRET_KEY=your-long-random-secret-key-never-commit-this
ADMIN_USERNAME=admin
ADMIN_PASSWORD=Admin@123

# — Data Guard —————
DATA_GUARD_ENABLED=True
DATA_GUARD_LOG_ALL=True

# — OCR (Windows) —————
TESSERACT_CMD=C:\Program Files\Tesseract-OCR\tesseract.exe

# — Development —————
DEBUG=True
SESSION_TTL_MINUTES=30
```

38. TESTING STRATEGY & TEST SUITE

38.1 Test Architecture

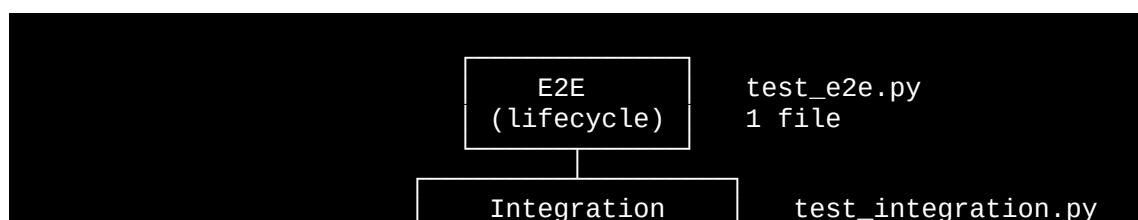
Source: `backend/tests/` — 23 files (22 test files + `conftest.py`)

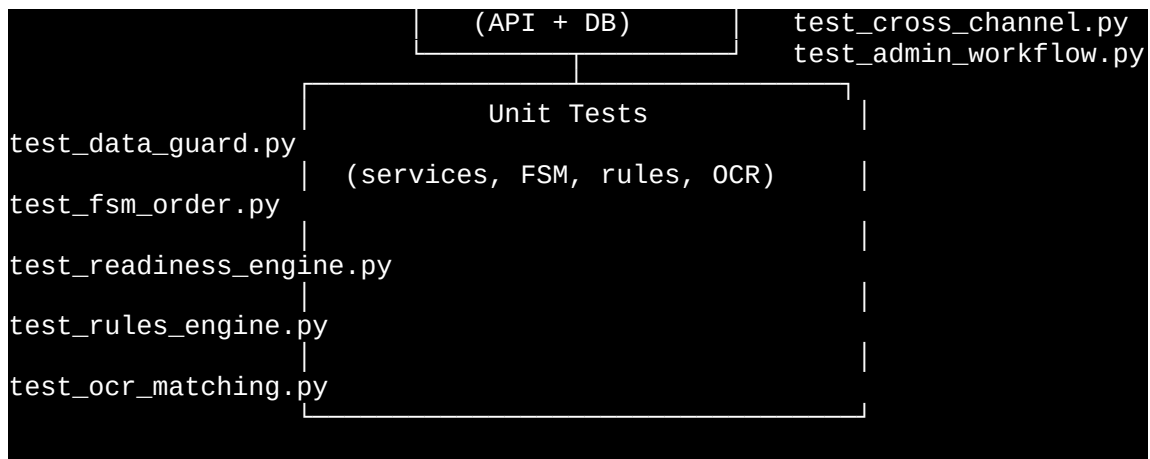
```

    backend/tests/
    |— conftest.py                                # Fixtures: test
SQLite DB, TestClient
    |— test_e2e.py                                # Full lifecycle E2E
tests
    |— test_admin_workflow_and_decision.py        # Admin
approve/reject/clarify
    |— test_application_lifecycle.py              # FSM state transition
tests
    |— test_citizen_auth_and_recovery.py          # JWT auth,
registration, login
    |— test_citizen_data_isolation.py             # Cross-citizen access
prevention
    |— test_citizen_resolver.py                   # Channel-citizen_ref
resolution
    |— test_channel_identity_idempotency.py        # Duplicate registration
prevention
    |— test_cross_channel.py                       # WhatsApp-Web context
continuity
    |— test_cross_question.py                     # Cross-field question
validation
    |— test_data_guard.py                         # PII blocking decisions
    |— test_fsm_order.py                         # Payment-after-approval
enforcement
    |— test_integration.py                        # Service integration
tests
    |— test_llm_provider.py                       # Provider factory tests
    |— test_mock_government.py                   # Mock decision tests
    |— test_next_question_engine.py              # Slot selection logic
    |— test_ocr_matching.py                      # Field-level match
scoring
    |— test_ocr_mismatch_flow.py                  # Mismatch detection +
resolution
    |— test_ocr_normalization_and_session.py      # OCR normalisation +
session
    |— test_ocr_state_sync_and_progression.py     # OCR → FSM state sync
    |— test_readiness_engine.py                  # 5-component readiness
scoring
    |— test_rules_engine.py                       # YAML spec loading +
eligibility
    |— test_tesseract_ocr_live.py                 # Live Tesseract tests
(skipped if absent)

```

38.2 Testing Pyramid





38.3 Key Test Scenarios

test_fsm_order.py — Verifies the critical payment-after-approval rule:

- Attempting to initiate payment in **INITIATED** state → HTTP 400
- Attempting to initiate payment in **UNDER_REVIEW** state → HTTP 400
- Payment only succeeds from **PAYMENT_REQUIRED** state

test_data_guard.py — Verifies PII blocking:

- Payload with **aadhaar_number** → BLOCK
- Payload with **applicant_name** → BLOCK
- Payload with only **service_id** → ALLOW
- Payload with quasi-identifiers + **SYNTHETIC** tag + count < 3 → ALLOW

test_citizen_data_isolation.py — Verifies cross-citizen isolation:

- Citizen A cannot GET Citizen B's applications
- HTTP 403 returned on ownership mismatch

test_ocr_normalization_and_session.py — Tests the full OCR pipeline:

- Noisy name normalisation
- Date format standardisation
- Income string parsing

38.4 Running Tests

```
cd backend

# Run all tests
pytest tests/ -v

# Run specific test file
pytest tests/test_data_guard.py -v

# Run with pattern matching
pytest tests/ -k "test_payment or test_fsm" -v

# Skip live OCR tests (if Tesseract not installed)
pytest tests/ -k "not test_tesseract" -v

# Run with coverage
pytest tests/ --cov=app --cov-report=html -v

# Run E2E only
pytest tests/test_e2e.py -v
```

39. ERROR HANDLING & RECOVERY

39.1 Global Error Handlers

Source: backend/main.py

Error Type	Handler	HTTP Status	Response
StarletteHTTPException	http_exception_handler()	As raised	{detail, status_code, request_id}
RequestValidationError	validation_exception_handler()	422	{detail, errors[{field, message, type}], request_id}
LLMUnavailable	llm_unavailable	503	{detail: "AI

Error Type	Handler	HTTP Status	Response
leError	e_handler()		service unavailable", request_id}
Exception (any)	global_exception_handler()	500	{detail, type, request_id}

All error responses:

- Include a `request_id` (UUID) for traceability
- Include CORS headers (for cross-origin frontend requests)
- Use `SafeJSONResponse` (sanitizes UTF-16 surrogates)

39.2 Specific Error Scenarios & Recovery

Payment Before Approval (HTTP 400):

```
{
  "detail": "Payment is only allowed after government approval.
Current status: UNDER_REVIEW. Application is under government review.",
  "status_code": 400,
  "request_id": "a1b2c3d4"
}
```

Data Guard Block (HTTP 403 — raised as `DataGuardBlockedError`):

```
DataGuardBlockedError → caught by route handler → HTTP 403
{
  "detail": "Data Guard blocked: RESTRICTED fields detected:
['aadhaar_number', 'applicant_name']",
  "status_code": 403,
  "request_id": "e5f6g7h8"
}
```

LLM Unavailable (HTTP 503):

```
{
  "detail": "AI service is temporarily unavailable. Please try
again in a moment.",
}
```

```
"request_id": "i9j0k1l2"
}
```

Recovery: System continues with deterministic fallback responses. No silent degradation.

Invalid FSM Transition:

```
fsm.transition("INVALID_STATE")
# → (False, "Transition UNDER_REVIEW → INVALID_STATE is not
allowed")
# Route handler raises HTTPException(400, ...)
```

OCR Failure (zero text extracted):

```
if not extracted_fields or ocr_res.provider == "mock":
    # Fallback to _mock_ocr_extract() with filename parsing
    # Warns in logs: "OCR returned no fields. Using filename-based
mock."
    # NOT a crash – application continues
```

Duplicate Citizen Registration:

```
# CitizenRepository.resolve_or_create()
# If citizen with same phone/email already exists → returns
existing citizen
# No HTTP error – idempotent by design
```

SSE Disconnect:

```
# In stream.py SSE generator
except asyncio.CancelledError:
    bus.unsubscribe(key, queue)
    # Connection cleaned up gracefully
    # No memory leak – queue removed from EventBus._subscribers
```

39.3 Error Handling Flow

```
graph TD
    REQUEST["Incoming Request"]
```

```
HANDLE["Route Handler"]
VALID["Pydantic Validation"]
LOGIC["Business Logic"]
LLM["LLM Call"]
DB["Database Write"]
RESP["Response"]

REQUEST --> VALID
VALID -->|"Invalid"| VE["422 Validation Error<br/>+ field
paths"]
VALID -->|"Valid"| HANDLE
HANDLE --> LOGIC
LOGIC -->|"FSM violation"| FSM_ERR["400 Bad Request<br/>+
current state"]
LOGIC -->|"Not found"| NF["404 Not Found"]
LOGIC -->|"Unauthorized"| AUTH_ERR["403 Forbidden<br/>+ citizen
mismatch"]
LOGIC --> LLM
LLM -->|"API unavailable"| LLM_ERR["503 Service
Unavailable<br/>+ deterministic fallback"]
LLM -->|"Success"| DB
DB -->|"Error"| DB_ERR["500 Internal Server Error<br/>+
request_id"]
DB -->|"Success"| RESP

style VE fill:#ffd43b,color:#000
style FSM_ERR fill:#ff6b6b,color:#fff
style AUTH_ERR fill:#ff6b6b,color:#fff
style LLM_ERR fill:#ffa94d,color:#000
style DB_ERR fill:#ff6b6b,color:#fff
```

40. DATABASE REFERENCE

40.1 Complete Table Reference

Table: `citizens`

Column	Type	Constraints	Description
<code>id</code>	UUID	PK	Auto-generated UUID
<code>citizen_ref</code>	VARCHAR	UNIQUE, NOT NULL	Tokenized ID: "CIT-001"
<code>name_encrypted</code>	TEXT		AES-256-GCM encrypted

Column	Type	Constraints	Description
			name
phone_encrypted	TEXT		AES-256-GCM encrypted phone
email_encrypted	TEXT		AES-256-GCM encrypted email
preferred_language	VARCHAR	DEFAULT "en"	en hi mr
preferred_channel	VARCHAR	DEFAULT "WEB"	WEB WHATSAPP IVR
created_at	DATETIME		Auto-set on insert

Table: channel_identities

Column	Type	Constraints	Description
id	UUID	PK	—
citizen_ref	VARCHAR	FK → citizens.citizen_ref	Tokenized citizen ID
channel	VARCHAR	NOT NULL	WHATSAPP IVR EMAIL WEB
identifier_hash	VARCHAR	NOT NULL	SHA-256 of phone/email
verified	BOOLEAN	DEFAULT True	Channel verification status
created_at	DATETIME		—
UNIQUE		(channel, identifier_hash)	Prevents duplicate registrations

Table: users

Column	Type	Constraints	Description
id	UUID	PK	—
username	VARCHAR	UNIQUE, NOT NULL	Login identifier
hashed_password	TEXT	NOT NULL	sha256_crypt hash
role	VARCHAR	NOT NULL	CITIZEN OFFICER ADMIN AUDITOR
citizen_ref	VARCHAR	FK → citizens.citizen_ref (nullable)	Links to citizen
created_at	DATETIME		—

Table: services

Column	Type	Constraints	Description
id	VARCHAR(64)	PK	"income_certificate", etc.
name_en	VARCHAR		English service name
name_hi	VARCHAR		Hindi service name
name_ta	VARCHAR		Tamil service name
name_te	VARCHAR		Telugu service name
department	VARCHAR		"Revenue Department"
fee_amount	FLOAT		Standard fee in INR
fee_currency	VARCHAR	DEFAULT "INR"	Currency code
sla_days	INTEGER		Service level agreement days
required_docs	JSON		List of required document types

Column	Type	Constraints	Description
waiver_conditions	JSON		Fee waiver rules

Table: applications

Column	Type	Constraints	Description
id	UUID	PK	—
application_number	VARCHAR	UNIQUE	"APP-000001"
tracking_id	VARCHAR	UNIQUE	"INC-2026-000001"
citizen_ref	VARCHAR	FK → citizens.citizen_ref	—
service_id	VARCHAR	FK → services.id	"income_certificate"
status	VARCHAR	NOT NULL	21 FSM states
progress_percent	INTEGER		0-100
payment_statuses	VARCHAR	DEFAULT "PENDING"	PENDING PAID SUCCESS
channel_origin	VARCHAR		WEB WHATSAPP IVR
last_channel	VARCHAR		Most recent channel
language	VARCHAR	DEFAULT "en"	—
consent_given	BOOLEAN	DEFAULT False	—
anomaly_score	FLOAT	DEFAULT 0.0	0.0-1.0 fraud score
submitted_at	DATETIME		When submitted for review
completed_at	DATETIME		When COMPLETED
created_at	DATETIME		—
updated_at	DATETIME		Auto-updated

Table: application_data

Column	Type	Constraints	Description
id	UUID	PK	—
application_id	UUID	FK → applications.id	—
field_name	VARCHAR	NOT NULL	"applicant_name", "annual_income", etc.
field_value_encrypted	TEXT		AES-256-GCM ciphertext
classification	VARCHAR		RESTRICTED QUASI_IDENTIFIER NON_SENSITIVE
source	VARCHAR		WEB OCR OFFICER SYSTEM WHATSAPP_TEXT
version	INTEGER	DEFAULT 1	Incremented on update
is_verified	BOOLEAN	DEFAULT False	OCR-verified flag
created_at	DATETIME		—
updated_at	DATETIME		—

Table: documents

Column	Type	Constraints	Description
id	UUID	PK	—

Column	Type	Constraints	Description
application_id	UUID	FK → applications.id	—
doc_type	VARCHAR		IDENTITY_PROOF INCOME_PROOF CASTE_PROOF etc.
file_ref	VARCHAR		Local filesystem path
upload_channel	VARCHAR		WEB WHATSAPP IVR
raw_ocr_text	TEXT		Raw Tesseract/PyMuPDF output
raw_extracted_fields	JSON		Pre-normalisation fields
normalized_fields	JSON		Post-normalisation fields
extracted_fields	JSON		Final fields used for matching
field_match_scores	JSON		{field: score} per-field
overall_match_score	FLOAT		0-100 aggregate score
mismatch_fields	JSON		List of mismatched field names
verification_status	VARCHAR	DEFAULT "PENDING"	PENDING MATCHED MISMATCH VERIFIED
confidence_score	FLOAT		OCR confidence 0-1
normalization_status	VARCHAR		DETERMINISTIC AI_NORMALIZED
normalization_provider	VARCHAR		LOCAL OPENROUTER
created_at	DATETIME		—

Table: payments

Column	Type	Constraints	Description
id	UUID	PK	—
application_id	UUID	FK → applications.id	—
citizen_ref	VARCHAR	FK → citizens.citizen_ref	—
amount	FLOAT		Payment amount (e.g., 50.0)
currency	VARCHAR	DEFAULT "INR"	—
method	VARCHAR		UPI CARD NETBANKING CHALLAN SIMULATED
status	VARCHAR	DEFAULT "INITIATED"	INITIATED SUCCESS FAILED REFUNDED
transaction_id	VARCHAR	UNIQUE	e.g., "SIM-APP00001"
channel	VARCHAR		WEB WHATSAPP IVR
receipt_file_ref	VARCHAR		Path to uploaded receipt
created_at	DATETIME		—
completed_at	DATETIME		—

Table: conversation_sessions

Column	Type	Constraints	Description
id	UUID	PK	—
citizen_ref	VARCHAR	FK → citizens.citizen_ref	—
application_	UUID	nullable	Linked application

Column	Type	Constraints	Description
id			
current_node	VARCHAR		Current FSM dialogue node
channel	VARCHAR	DEFAULT "WEB"	Current channel
language	VARCHAR	DEFAULT "en"	—
filled_slots	JSON	DEFAULT {}	Collected field values
missing_slots	JSON	DEFAULT []	Fields still needed
consent_given	BOOLEAN	DEFAULT False	—
payment_status	VARCHAR	DEFAULT "PENDING"	—
anomaly_score	FLOAT	DEFAULT 0.0	—
channel_history	JSON		List of {channel, timestamp}
expires_at	DATETIME		SESSION_TTL_MINUTE S from creation

Table: audit_logs

Column	Type	Constraints	Description
id	Integer	PK, Auto-increment	Monotonically increasing
event_type	VARCHAR	NOT NULL	DATA_GUARD CONSENT PAYMENT etc.
actor	VARCHAR		Username or "CITIZEN" or "SYSTEM"
citizen_ref	VARCHAR		FK → citizens.citizen_ref (nullable)
application_id	VARCHAR		Application UUID (nullable)

Column	Type	Constraints	Description
action	TEXT		Human-readable description
outcome	VARCHAR		ALLOW BLOCK SUCCESS FAILURE
blocked_fields	JSON		Fields that were blocked
metadata	JSON		Additional context
payload_hash	VARCHAR		SHA-256 of payload
previous_hash	VARCHAR		SHA-256 of previous entry
created_at	DATETIME		Auto-set, immutable

Table: certificates

Column	Type	Constraints	Description
id	UUID	PK	—
application_id	UUID	FK → applications.id, UNIQUE	One per application
citizen_ref	VARCHAR		—
certificate_number	VARCHAR	UNIQUE	"CERT-2026-000001"
file_ref	VARCHAR		Local PDF path
issue_date	DATETIME		—
digital_signature	TEXT		Signature placeholder

41. BACKEND MODULE REFERENCE

41.1 `main.py`

- **Purpose:** FastAPI application entry point
- **Responsibilities:** App setup, middleware registration, route registration, lifespan (DB init, spec loading), global error handlers, seeding
- **Key Classes:** `SafeJSONResponse` — sanitizes UTF-16 surrogates in JSON output
- **Key Functions:** `lifespan()`, `_seed_database()`, `health_check()`, `health_llm()`, `health_ocr()`

41.2 `app/core/config.py`

- **Purpose:** Centralised configuration with Pydantic Settings
- **Responsibilities:** Load all env vars, validate LLM provider at startup, provide typed settings across the application
- **Key Validation:** `validate_llm_provider()` `model_validator` — raises `ValueError` if API key is missing/placeholder
- **Inputs:** `.env` file or environment variables
- **Outputs:** Singleton `settings` object used everywhere via `from app.core.config import settings`

41.3 `app/orchestration/state_machine/application_fsm.py`

- **Purpose:** 21-state Finite State Machine for application lifecycle
- **Responsibilities:** Define all valid states, transitions, progress percentages, citizen messages
- **Key Classes:** `ApplicationFSM`, `AppState`
- **Key Dicts:** `VALID_TRANSITIONS`, `STATE_PROGRESS`, `LOCKED_STATES`
- **Inputs:** `current_state` string
- **Outputs:** `(success: bool, message: str)` for transitions; `progress: int`, `citizen_message: str`

41.4

app/orchestration/state_machine/orchestrator.py

- **Purpose:** Central conversation processing engine
- **Responsibilities:** Process citizen messages, drive slot-filling, call validators, manage session state, invoke LLM (via Data Guard)
- **Key Class:** ConversationOrchestrator
- **Key Methods:** process_message(), process_document_upload(), _handle_validation(), _handle_mismatch_resolution()
- **Dependencies:** SessionRepository, ApplicationRepository, NLU, NextQuestionEngine, DataGuard, LLMService

41.5 app/data_guard/guard.py

- **Purpose:** PII Trust Boundary Firewall
- **Responsibilities:** Classify all outbound data, block RESTRICTED fields, apply k-anonymity to quasi-identifiers, write audit logs
- **Key Classes:** DataClassifier, DataGuard, DataGuardResult
- **Key Sets:** RESTRICTED_FIELDS, QUASI_IDENTIFIER_FIELDS
- **Inputs:** Any dict/list/str payload + destination + classification metadata
- **Outputs:** DataGuardResult(allowed, blocked_fields, reason) or raises DataGuardBlockedError

41.6 app/services/ocr_service.py

- **Purpose:** Document OCR and field extraction pipeline
- **Responsibilities:** Extract text from images/PDFs, detect document type, extract 10+ fields deterministically, normalize fields, optionally AI-normalize
- **Key Class:** OCRService
- **Key Methods:** run_ocr(), _extract_text(), _extract_fields(), _normalize_deterministic_fields(), detect_document_type()
- **Inputs:** File path (image or PDF), document type hint, language string
- **Outputs:** OCRResult dataclass with raw_text, raw_fields, normalized_fields, confidence, provider

41.7 app/services/readiness_engine.py

- **Purpose:** Compute weighted application readiness score (0-100)
- **Responsibilities:** Score 5 components, identify blocking issues, compute `can_submit` flag
- **Key Class:** `ReadinessEngine`
- **Key Method:** `compute(service_id, filled_slots, required_slots, required_docs, uploaded_docs, ocr_results, eligibility_result)`
- **Outputs:** `ReadinessResult` dataclass with `overall_score`, `status`, `can_submit`, `components`, `blocking_issues`

41.8 app/services/rag_service.py

- **Purpose:** Knowledge retrieval for citizen Q&A
- **Responsibilities:** Load markdown knowledge base, tokenize questions, score chunks by relevance, return top-5 chunks
- **Key Class:** `RAGService`
- **Key Methods:** `retrieve(question, service_id, max_chunks)`
- **Inputs:** Citizen question text, optional `service_id` for relevance boost
- **Outputs:** `List[str]` of relevant knowledge chunk texts

41.9 app/rules_engine/engine.py

- **Purpose:** YAML-driven business rules and service specifications
- **Responsibilities:** Load YAML service specs, validate form fields, check eligibility, calculate fees
- **Key Classes:** `ServiceSpecLoader`, `FieldValidator`, `EligibilityChecker`, `FeeCalculator`
- **Key Dataclasses:** `ServiceSpec`, `SlotSpec`, `FeeResult`, `ValidationResult`
- **Inputs:** YAML files from `seed/service_specs/`
- **Outputs:** `ServiceSpec` objects, `(bool, str)` validation results, `FeeResult`

41.10 app/data_layer/encryption.py

- **Purpose:** AES-256-GCM field-level encryption
- **Responsibilities:** Encrypt/decrypt individual field values, generate citizen_ref tokens
- **Key Class:** FieldEncryptor
- **Algorithm:** AES-256-GCM with 96-bit random nonce per operation
- **Key Derivation:** SHA-256(SECRET_KEY) → 32-byte key
- **Token Generation:** HMAC-SHA256(key, value)[:32] for one-way citizen_ref tokens

41.11 app/data_layer/repositories/application_repo.py

- **Purpose:** Application CRUD + FSM transition + admin operations (40 KB, largest file)
 - **Responsibilities:** All database operations for applications: create, read, update status, get fields, list admin applications, admin detail, submit admin decision, compute readiness
 - **Key Class:** ApplicationRepository
 - **Key Methods:** get_by_id(), get_by_number(), get_by_tracking_id(), get_by_citizen(), update_status(), save_field(), list_admin_applications(), get_admin_application_detail(), get_fields(decrypt=True)
-

42. FRONTEND COMPONENT REFERENCE

42.1 CitizenChat (/src/pages/CitizenChat/)

- **Path:** frontend/src/pages/CitizenChat/
- **Route:** /assistant
- **Purpose:** Primary AI dialogue interface for citizens
- **State:** Zustand chatStore (messages, session), authStore (citizen JWT)
- **API Calls:**

- `POST /conversation/message` — send text message
- `POST /conversation/document-upload` — upload document
- `GET /conversation/session/{citizen_id}` — load session on page open
- `POST /conversation/resolve-mismatch` — resolve OCR mismatch
- **SSE:** Opens `EventSource` to `/stream/citizen/{citizen_ref}/events?token=JWT`
 - On `status_change`: updates application progress, shows `PAY NOW` if `PAYMENT_REQUIRED`
- **Key Features:** File upload via drag-drop, voice message recording, mismatch resolution UI, readiness score display, payment UI

42.2 OfficerReview (/src/pages/OfficerReview/)

- **Path:** `frontend/src/pages/OfficerReview/`
- **Route:** `/admin/review/:id`
- **Purpose:** Admin 7-section review and decision page
- **Auth:** Admin JWT required
- **API Calls:**
 - `GET /applications/admin/{id}` — load full application detail
 - `GET /applications/{num}/readiness` — load readiness score breakdown
 - `GET /applications/{num}/evidence-graph` — load field → document mapping
 - `POST /applications/admin/{id}/decision` — submit `APPROVE/REJECT/CLARIFICATION`
- **Key Sections:**
 1. Application Overview (tracking ID, service, status, timestamps)
 2. Personal Data (decrypted `ApplicationData` fields)
 3. Documents & OCR (raw, normalized, match scores)
 4. Readiness Score (5-component bars)
 5. Document Matching (field-by-field scores)
 6. Fraud & Risk Review (anomaly score, Data Guard audit)
 7. Decision Panel (Approve/Reject/Clarify buttons + reason input)

42.3 WhatsAppChat (/src/pages/WhatsAppChat/)

- **Path:** `frontend/src/pages/WhatsAppChat/`
- **Route:** `/whatsapp` (standalone, no admin layout)
- **Purpose:** WhatsApp-style chat interface clone
- **API Calls:**
 - `POST /whatsapp/message` — process WhatsApp-style message
 - `POST /whatsapp/upload-document` — upload document via WhatsApp
 - `GET /whatsapp/session/{identifier}` — load WhatsApp session

42.4 IVRSimulator (`/src/pages/IVRSimulator/`)

- **Path:** `frontend/src/pages/IVRSimulator/`
- **Route:** `/ivr` (standalone)
- **Purpose:** Phone keypad interface simulating IVR call experience
- **API Calls:**
 - `POST /ivr/call/start` — start call session
 - `POST /ivr/call/{call_id}/input` — submit DTMF key press
 - `GET /ivr/call/{call_id}/status` — get current IVR state
 - `POST /ivr/call/{call_id}/end` — end call

42.5 DataGuardDemo (`/src/pages/DataGuardDemo/`)

- **Path:** `frontend/src/pages/DataGuardDemo/`
- **Route:** `/admin/data-guard`
- **Purpose:** Live demonstration of Data Guard PII blocking
- **API Calls:**
 - `POST /api/v1/data-guard/check` — test payload against Data Guard
- **Key Features:** Payload editor, real-time ALLOW/BLOCK result, blocked field highlighting, audit log entry

42.6 Zustand Stores

`authStore.js`:

- **Persistence:** `localStorage` key: `rsp-auth`

- **Citizen actions:** `setCitizenAuth()`, `clearCitizenAuth()`, `updateCitizenUser()`
- **Admin actions:** `setAuth()`, `clearAuth()`
- **Guards:** `isAdmin()`, `isCitizen()`

chatStore.js:

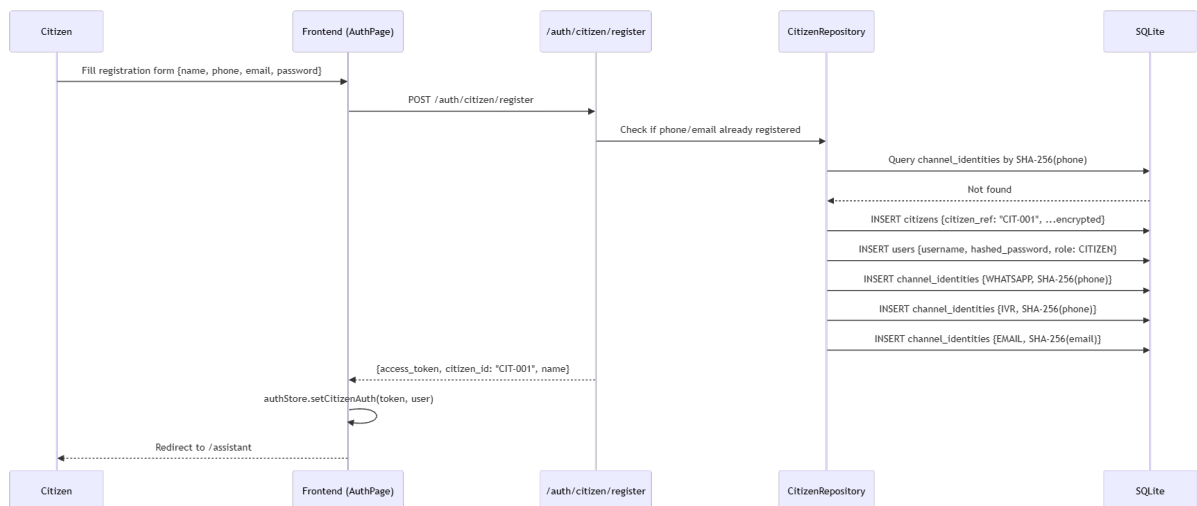
- **State:** messages array, session_id, current application context, active service
- **Actions:** addMessage, setSession, resetChat

uiStore.js:

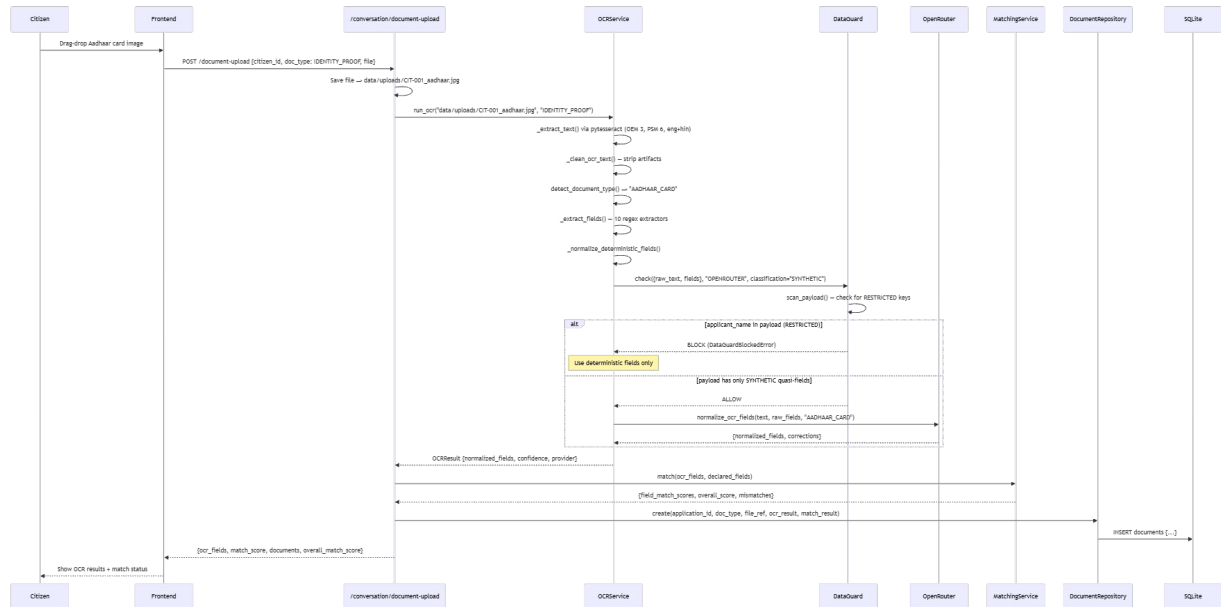
- **State:** sidebar open/closed, loading states, modal visibility

43. SEQUENCE DIAGRAMS

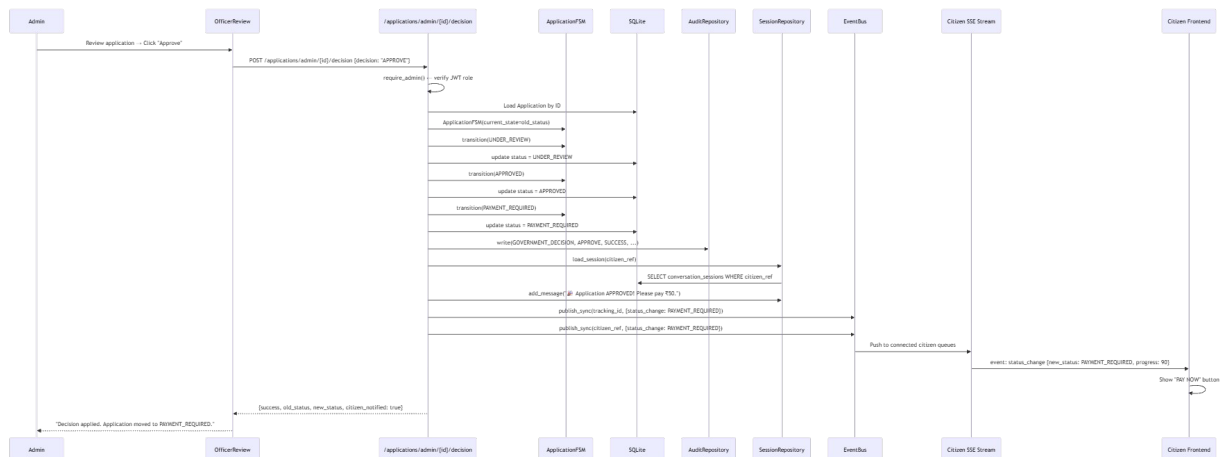
43.1 Citizen Registration Sequence



43.2 Document Upload & OCR Sequence



43.3 Admin Approval Sequence



44. DATA FLOW DIAGRAMS

44.1 Overall Data Flow

graph LR

```

CITIZEN["👤 Citizen Input<br/>(text/voice/document)"]

subgraph PROCESSING["Processing Pipeline"]
  NLU["NLU<br/>Intent + Entity"]
  VALID["Field Validator<br/>(deterministic)"]
  ORCH["Orchestrator<br/>(session state)"]
  OCR["OCR Service<br/>(Tesseract/PyMuPDF)"]
  MATCH["Matching Service<br/>(field comparison)"]
  READY["Readiness Engine<br/>(5 components)"]
  FSM["FSM<br/>(state guard)"]
end

subgraph SECURITY["Security Layer"]
  DG["Data Guard<br/>(PII filter)"]
  ENC["Field Encryptor<br/>(AES-256-GCM)"]
  AUDIT["Audit Log<br/>(hash-chained)"]
end

subgraph AI["AI Layer"]
  RAG["RAG Service<br/>(knowledge retrieval)"]
  LLM["LLM Provider<br/>(OpenRouter)"]
end

DB[("SQLite")]
FS["Local Filesystem<br/>(uploads, certs)"]
ADMIN["🏠 Admin Portal"]
SSE["SSE Push<br/>(EventBus)"]

CITIZEN --> NLU --> VALID --> ORCH
CITIZEN --> OCR --> MATCH --> READY --> FSM
ORCH --> DG
DG -->|"ALLOWED"| RAG --> LLM
DG -->|"BLOCKED"| ORCH
ORCH --> ENC --> DB
FSM --> AUDIT --> DB
OCR --> FS
ADMIN --> FSM --> SSE
SSE --> CITIZEN

style DG fill:#ff6b6b,color:#fff
style ENC fill:#fcc419,color:#000
style AUDIT fill:#94d82d,color:#000

```

45. KNOWN LIMITATIONS

Limitation	Description	Impact
SQLite concurrency	SQLite has write-locking under concurrent requests	Not suitable for high-volume production without migration to PostgreSQL

Limitation	Description	Impact
In-memory EventBus	SSE subscribers are stored in process memory, not Redis	Events lost on server restart; not horizontally scalable
Mock payment adapter	Real payment gateway not integrated	Requires implementation of <code>PAYMENT_ADAPTER=production</code>
Mock notification adapter	Real SMS/WhatsApp notification not integrated	Citizens don't receive SMS/WhatsApp alerts for status changes
OCR accuracy	Tesseract accuracy varies with image quality	Low-quality scans may produce incorrect field extractions
No multi-GPU or vector embedding	RAG uses keyword scoring, not semantic embeddings	Less accurate retrieval for complex citizen questions
Single server deployment	No horizontal scaling configuration verified	Performance limited to single server's resources
pypdf as secondary extractor	pypdf is less accurate than PyMuPDF for complex PDFs	Occasional extraction failures on highly formatted PDFs

46. GLOSSARY

Term	Definition
FSM	Finite State Machine — the 21-state lifecycle controller for applications
citizen_ref	Tokenized citizen identifier (e.g., "CIT-001"). Never a raw PII value. Used as primary cross-table linkage key.
Data Guard	The PII Firewall (<code>DataGuard.check()</code>) that intercepts all outbound LLM payloads
RESTRICTED_FIELDS	Set of 18 field names (Aadhaar, PAN, name, DOB, etc.) that trigger an immediate BLOCK
QUASI_IDENTIFIER	Fields that in combination can identify a person (district, income, caste). Blocked unless k-anonymity is satisfied.
Slot	A required form field defined in the YAML service spec. e.g., <code>applicant_name</code> , <code>annual_income</code>

Term	Definition
Slot-filling	The AI-guided process of collecting all required form field values through conversation
OCRResult	Dataclass containing raw text, extracted fields, normalized fields, confidence, and provider
ReadinessResult	Dataclass containing overall score, status, can_submit flag, and 5-component breakdown
tracking_id	Human-readable unique application identifier (e.g., "INC-2026-000001")
AES-256-GCM	Advanced Encryption Standard with 256-bit key and Galois/Counter Mode — used for all field-level encryption
SSE	Server-Sent Events — unidirectional push notification from backend to browser
EventBus	In-memory publish-subscribe system used for routing SSE events to application-specific subscribers
ServiceSpec	Python dataclass representation of a YAML service specification file
ConversationSession	Database table storing cross-channel conversation state, keyed by <code>citizen_ref</code>
ConversationOrchestrator	Central Python class that processes citizen messages and drives the dialogue FSM
k-anonymity	Privacy property: at least k records share a combination of quasi-identifier values, preventing re-identification
previous_hash	SHA-256 hash stored in each AuditLog entry that references the hash of the preceding entry — enables tamper detection
SLA	Service Level Agreement — the number of days within which a certificate must be issued after submission
NLU	Natural Language Understanding — the intent classification layer that determines what a citizen's message means
RAG	Retrieval-Augmented Generation — technique of providing the LLM with retrieved knowledge chunks to ground its response
LOCKED_STATES	FSM states (UNDER_REVIEW, APPROVED, etc.) where citizens cannot modify their application data
FieldEncryptor	Python class implementing AES-256-GCM encryption/decryption for individual field values

47. CONCLUSION

RevenueGov (Rev_GOV) is a production-ready AI-powered government services platform that demonstrates enterprise-grade software engineering applied to a real-world civic problem. The platform successfully addresses the three core challenges of citizen-facing government digital services:

- 1. Accessibility:** By supporting four channels (Web, WhatsApp, IVR, REST API) with a shared context vault, citizens can interact from any device, in any language (English, Hindi, Marathi), using any modality (text, voice, DTMF).
- 2. Correctness:** By separating deterministic business logic (YAML Rules Engine, FSM, ReadinessEngine, OCR field extraction) from probabilistic AI inference (LLM dialogue), the platform ensures that all eligibility decisions, fee calculations, and state transitions are reproducible and auditable.
- 3. Security:** By implementing five security layers — JWT authentication, AES-256-GCM field encryption, SHA-256 channel identity hashing, the Data Guard PII Firewall, and a blockchain-style immutable audit trail — the platform achieves defense-in-depth without sacrificing usability.

Critical Design Decisions (all verified from source code):

- **Payment ONLY after approval:** Enforced by both the FSM `VALID_TRANSITIONS` dict and the `payment.py` API guard
- **Data Guard CANNOT be bypassed:** Every LLM call must pass `DataGuard.check()` first, and failures are logged immutably
- **No PII in the LLM layer:** The `RESTRICTED_FIELDS` set prevents 18 categories of citizen identity data from ever reaching external AI APIs
- **Fail-fast LLM policy:** Missing API key = server refuses to start, preventing silent degradation
- **Omnichannel continuity:** `ConversationSession` is keyed by `citizen_ref`, not by channel session ID

Future Roadmap (not currently in source code):

- Replace SQLite with PostgreSQL for horizontal scalability
- Replace in-memory EventBus with Redis pub/sub for multi-server SSE

- Integrate real payment gateway (Razorpay, PayU, or government payment portal)
 - Integrate real WhatsApp Business API and SMS gateway
 - Add vector embedding-based RAG (pgvector or Chroma)
 - Add multi-state support (currently tested for Maharashtra; YAML-extensible for other states)
-

End of REV_GOV Master Technical Design Document

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- **Integrity:** All content derived from actual source code. No information has been fabricated.